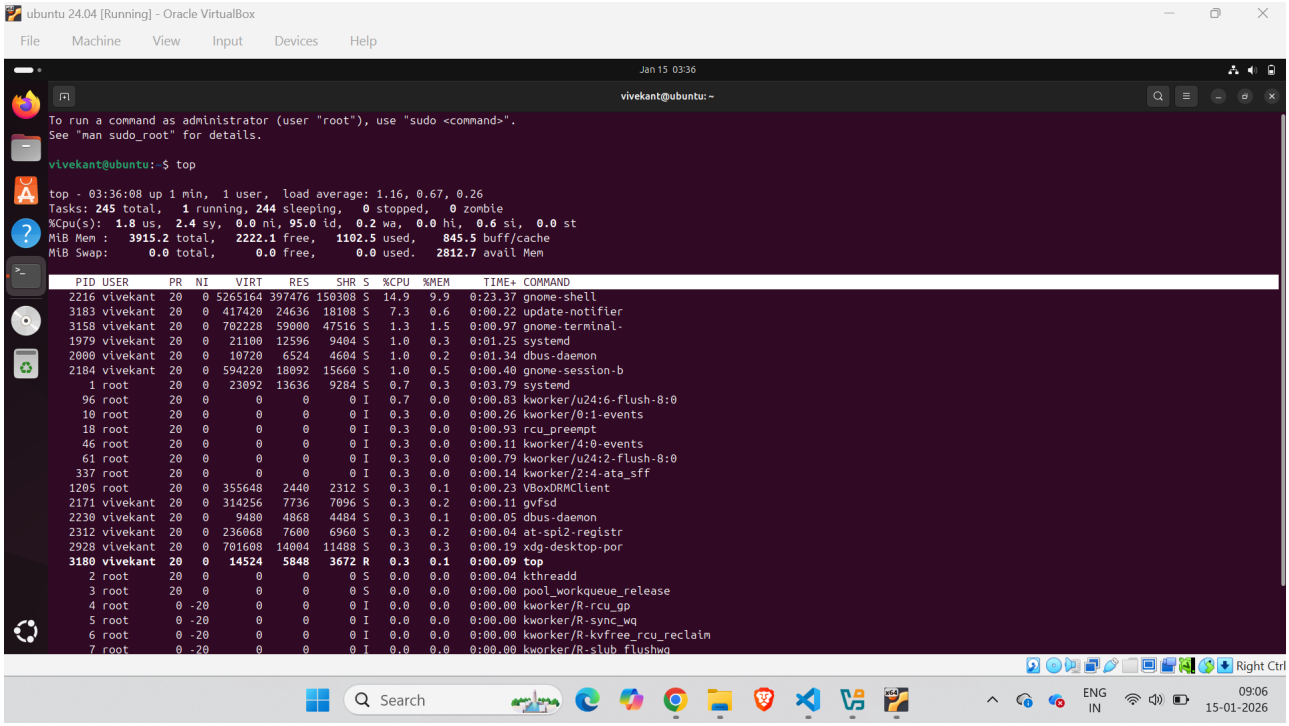
Linux commands

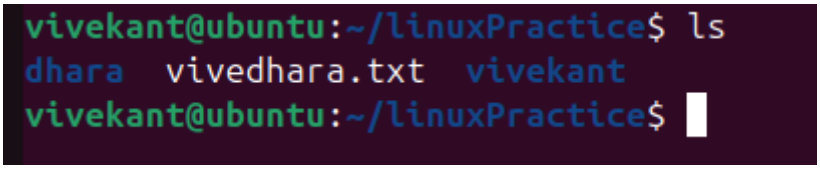
top

- status of your system
- like number of process , threads, which process is running/sleeping/killed/zombie, waken time of system , owner of process or thread, process ID(PID)



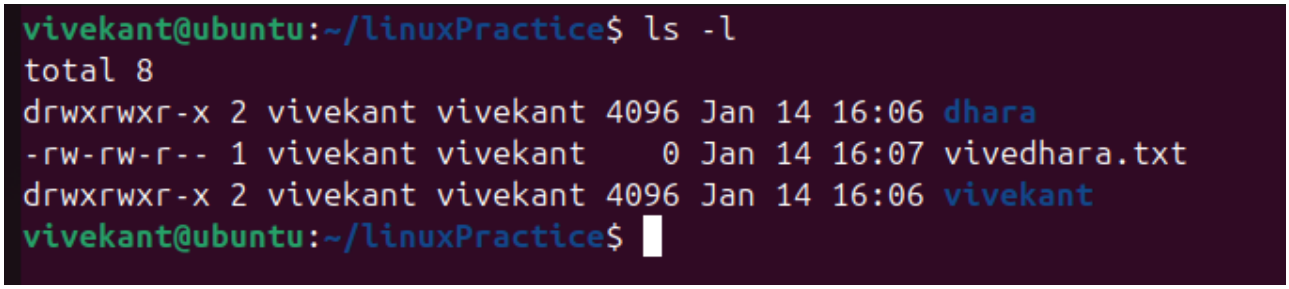
ls

- list of directory and files



ls -l

- list of directory and files with dr for directory and _ for files, you can see in image



pwd(present working directory)

cd

- to change directory

cd ..

- to go back to previous directory

touch filename.txt

- create filename.txt file in pwd

rm filename.txt

- to remove a single file from directory at once

rmdir vivekant/

- to remove empty directory only

rm -r vivekant/

- it remove all file from vivekant and remove directory vivekant too

echo "i will be billionare"

- to print somethign in shell

echo "dhara" > vivedhara.txt

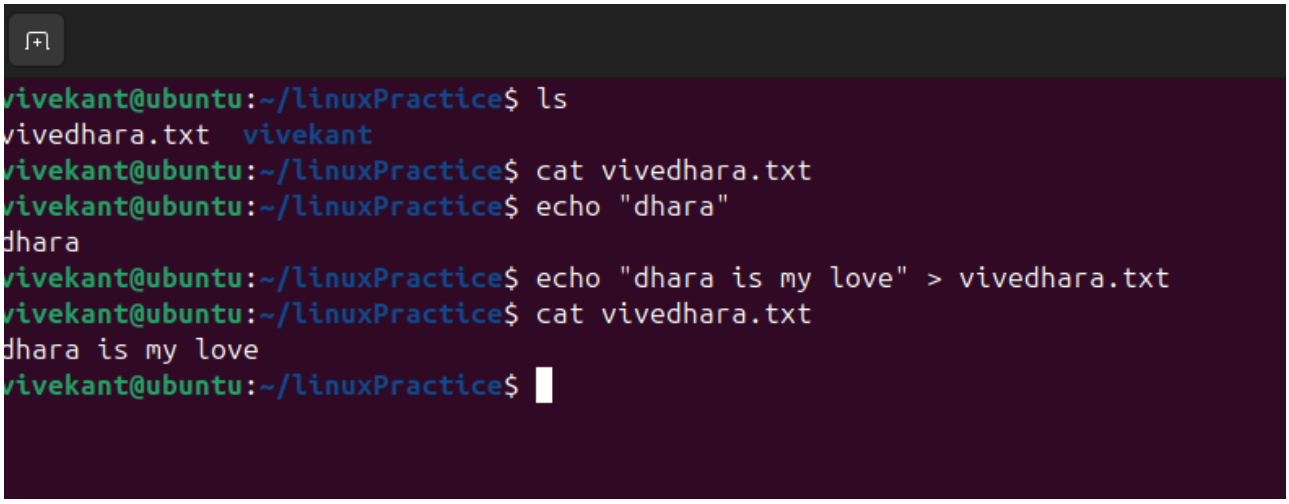
- this command will write dhara in vivedhara.txt file
- ">" it means redirect
- jo bhi echo ke samne daloge vo print kar dega
- ">" ye redirect kar deta hai like ki vivedhara.txt file me print kar de
- echo ka kaam hai print karna bas
- yadi file exist nahi karta toh autocreate kar deta hai

vim app.log(filename)

- given filename not necessary to exist
- after running this command , if you press 'i' then you can write whatever you want in file
- after writting click esc to escape from insert mode
- then if you want to save the changes , write :w and then enter
- if you want exit then write :q and press enter
- if you want save change and exit write :wq and then press enter

cat vivedhara.txt

- it display or print content inside vivedhara.txt



zcat vivedhara.txt

- it display or print content inside zipped vivedhara.txt

head vivedhara.txt

- print first 5 line of vivedhara.txt

tail vivedhara.txt

- print last 5 line of vivedhara.txt

tail -f vivedhara.txt

- if not work use -F instead of -f
- continuosely watching for last 5 line, if new line get added it will show new last five line

watch -n 5 tail -n 6 vivedhara.txt

- 'watch -n 5' means command next to it run after every 5 second
- its now working in your virutal box so dont confuse it
- 'tail -n 6 -F file.txt' behave same as it behave originally

less vivedhara.txt

- it is different from 'cat vivedhara.txt' because in cat command +enter program get exit and you just scross history of command output
- but in less command + enter you have poiter where you can do searching, smotth scroling and lots of thing, below are the command for perform action
- program are still running for exit the program we enter 'q'

- enter and down arrow feature are same

Key	Kya karta hai
`Space`	Next page
`b`	Previous page
`Enter`	1 line neeche
`↑ / ↓`	Line by line
`g`	File ka start
`G`	File ka end
`d`	Half page down
`u`	Half page up

cp source destination

- to copy file from souce to destination

```
cp vdss/file1.txt vivekant/    ----> file1.txt will get copied in vivekant folder
cp vdss/file1.txt vivekant/new_file.txt -----> file1.txt will get copied in
vivekant folder with name new_file.txt
```

cp -r source destination

- to copy folder with their file from source to destination

```
cp -r vdss/ vivekant/    ----> vdss folder will get copied in vivekant folder with
name vdss
cp -r vdss/ vivekant/dss -----> vdss folder  will get copied in vivekant  folder
with name dss
```

mv source destination

- move file or folder from source to destination
- this command also used for rename like following

```
mv vdss dss
so now vdss folder will rename as dss
```

wc vdss.txt

- word count
- 1=number of line, 4=number of words, 18= number of bites in vdss.txt file



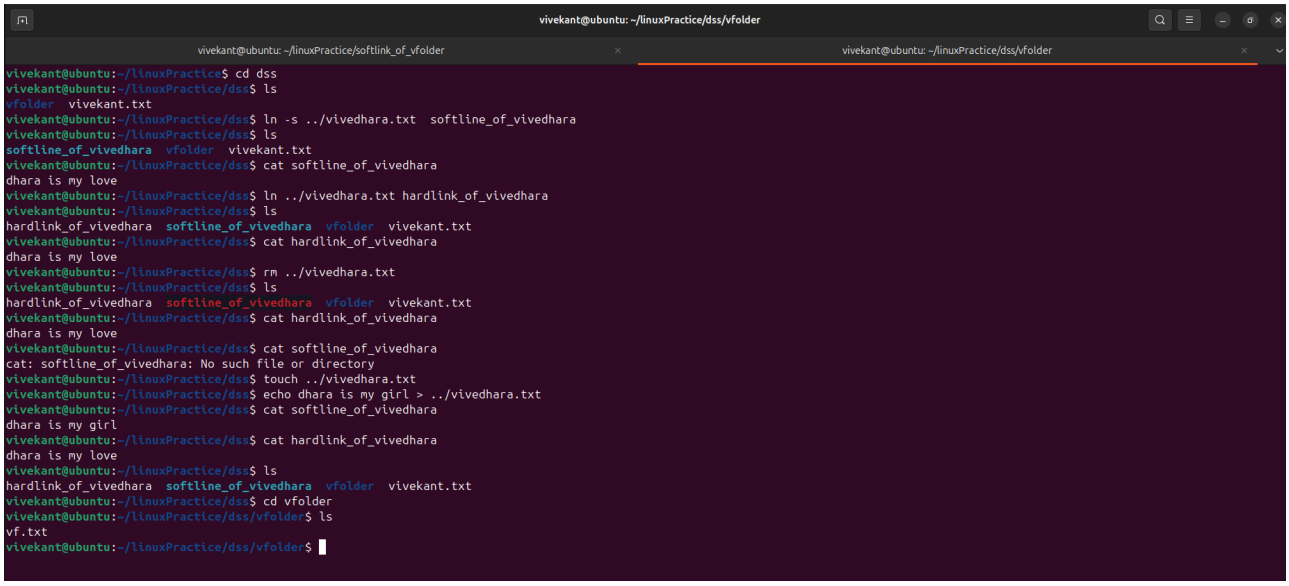
hardlink and softlink are shortcut of file or folder in a directory so that you can go that path directely by clicking on shortcut such that you dont have to navigage directory one by one

ln vivekant/dss.txt name_of_hardlink_of_file

- this is use for create hardlink of file
- hardlink only created for file not for folder
- even original file get deleted , you can view the file from hardlink

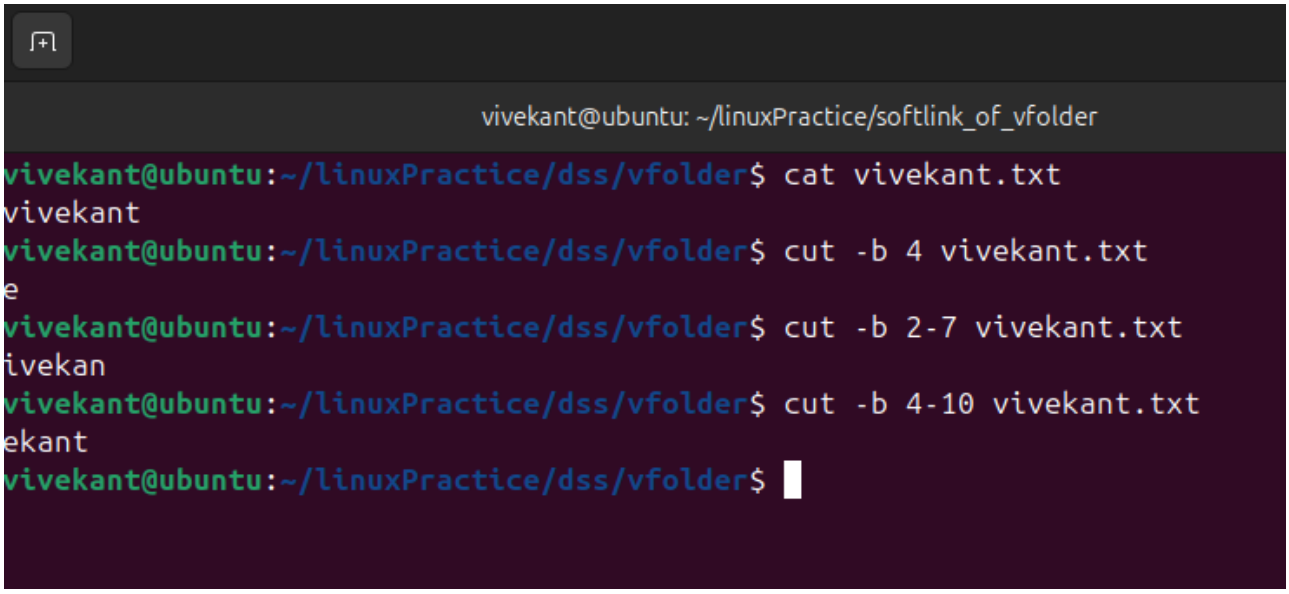
ln -s vivekant/dss.txt name_of_softlink_of_file

- this is use for create softlink of file or folder
- softlink created for both file and folder
- on deletion of original file or folder , you cant see or retrive file/folder from shortcut softlink
- on deletion color of softlink become red on ls command of parent directory



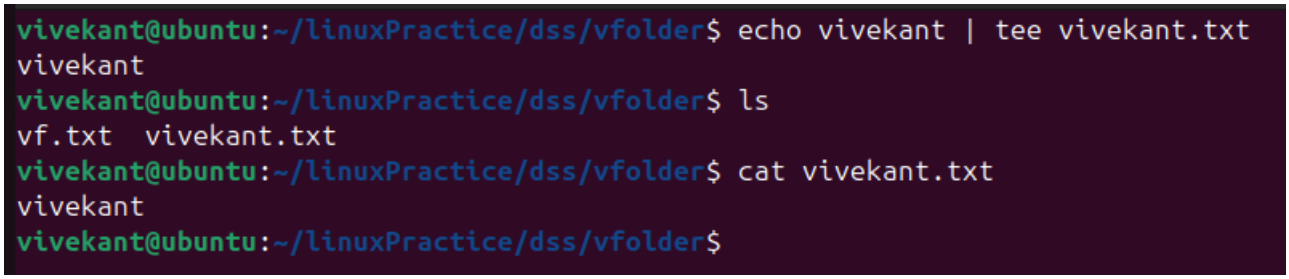
cut -b 1-4 vivedhara.txt

- it will give you 1-4 bite of data inside vivedhara.txt



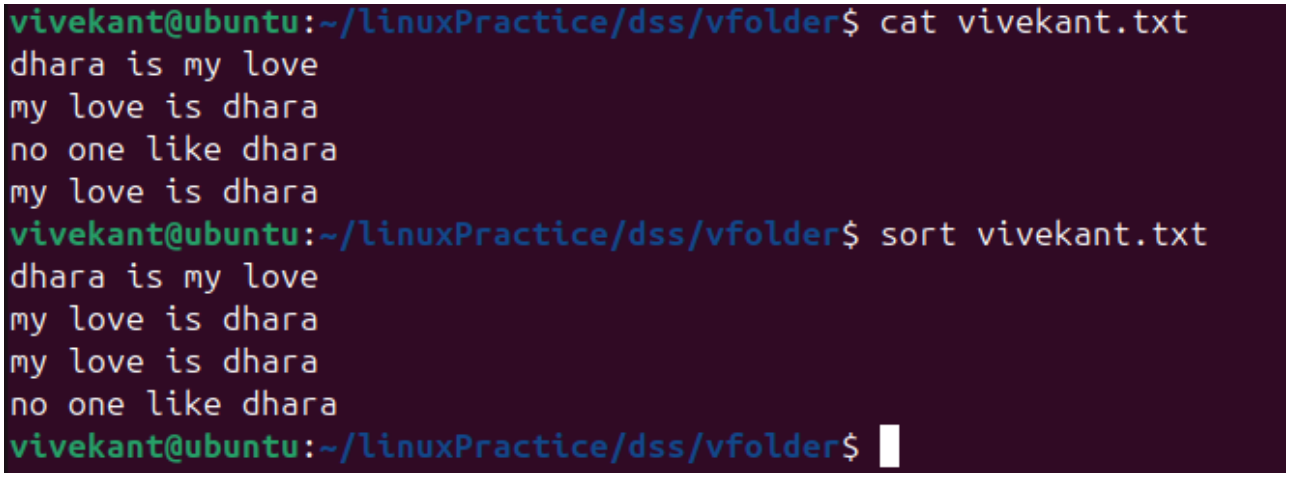
echo "dss is my love" | tee vivekant.txt

- it will do both display and write content in file simultaneously



sort vivedhara.txt

- it will sort in ascending order of content in vivedhara.txt line by line



diff file.txt file2.txt

- it will give you difference between two file

- it will print lines which are not in both file

```
vivekant@ubuntu:~/linuxPractice/dss/vfolder$ cat dhara.txt
dhara is mine
dhara is my love
vivekant@ubuntu:~/linuxPractice/dss/vfolder$ cat vivekant.txt
dhara is my love
my love is dhara
no one like dhara
my love is dhara
vivekant@ubuntu:~/linuxPractice/dss/vfolder$ diff vivekant.txt dhara.txt
0a1
> dhara is mine
2,4d2
< my love is dhara
< no one like dhara
< my love is dhara
vivekant@ubuntu:~/linuxPractice/dss/vfolder$
```

imp---> hidden folder name start with .folder_name

imp---> 1 byte = 8 bits

df | df -h | du | ls -a

- du file ka size nhi batata, bus folder ka size and total size of current directory(.)
- size kb mein hota hai
  - `ls -a` → folder ke andar **saari files dikhata hai**, hidden files bhi (jo `.` se start hoti hain)
  - `df` → system ke **disk partitions ka total aur free space** dikhata hai
  - `df -h` → `df` jaisa hi hai, bas **size human-readable format** mein dikhata hai (KB, MB, GB)
  - `du` → **current folder aur uske andar ki files/folders kitni disk space le rahe hain** wo dikhata hai

```
vivekant@ubuntu:~/linuxPractice$ ls
class.txt  dss  dss.txt  softlink_of_vfolder  vivedhara.txt  vivekant  vivekant.txt
vivekant@ubuntu:~/linuxPractice$ df
Filesystem     1K-blocks    Used Available Use% Mounted on
tmpfs             400916    1472    399444   1% /run
/dev/sda2       31636312 7460224  22543476  25% /
tmpfs            2004560       0   2004560   0% /dev/shm
tmpfs               5120       8      5112   1% /run/lock
tmpfs             400912     128    400784   1% /run/user/1000
/dev/sr0         6243054 6243054         0 100% /media/vivekant/Ubuntu 24.04.3 LTS amd64
vivekant@ubuntu:~/linuxPractice$ du
28      ./vivekant
40      ./dss/vfolder
52      ./dss
100     .
vivekant@ubuntu:~/linuxPractice$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           392M  1.5M  391M   1% /run
/dev/sda2        31G  7.2G   22G  25% /
tmpfs           2.0G     0  2.0G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           392M  128K  392M   1% /run/user/1000
/dev/sr0        6.0G  6.0G     0 100% /media/vivekant/Ubuntu 24.04.3 LTS amd64
vivekant@ubuntu:~/linuxPractice$ ls -a
.  ..  class.txt  dss  dss.txt  softlink_of_vfolder  vivedhara.txt  vivekant  vivekant.txt
vivekant@ubuntu:~/linuxPractice$
```

chmod 400 linuxfordevops.pem

- run this before start ssh connection

- it check this file accessibilty if private then ok otherwise ssh connect refuses by server itself

1 `chmod` = change mode (permissions)

Linux files have **3 permission sets:**

Who	Symbol
Owner (you)	`u`
Group	`g`
Others	`o`

And **3 permissions:**

Permission	Value
read	`4`
write	`2`
execute	`1`

---

2 Meaning of `400`

`400` is an **octal permission number:**

Digit	Applies to	Meaning
`4`	owner	read only
`0`	group	no access
`0`	others	no access

3 Why this is REQUIRED for SSH keys

Your `.pem` file is a **private key** (secret).

If:

- group or others can read it → **anyone could steal your key**

So SSH does this security check:

> "If this private key is readable by anyone except the owner, I will NOT use it."

If permissions are too open, SSH throws:

vbnet Copy code

```
WARNING: UNPROTECTED PRIVATE KEY FILE!
Permissions 0644 for 'linuxfordevops.pem' are too open.
```

- when you change the mode of identity file to write or execute from read then on ssh command following thing will happen
    1. mine pc will not allow ec2 to read identity file and give error permisson denied
    2. due to this erorr ec2 will give error connection request failed which is made by public key method

```
vivekant@ubuntu:~/linuxPractice$ chmod 200 linuxfordevops.pem
vivekant@ubuntu:~/linuxPractice$ ssh -i linuxfordevops.pem ubuntu@ec2-3-110-213-115.ap-south-1.compute.amazonaws.com
Load key "linuxfordevops.pem": Permission denied
ubuntu@ec2-3-110-213-115.ap-south-1.compute.amazonaws.com: Permission denied (publickey).
vivekant@ubuntu:~/linuxPractice$
```

ssh connection

- command

```
ssh -i key-pair-file-path username@public_ipv4_dns
ssh---> secure shell
-i ---> identity file which contains privaate key
username for aws by defalt ubuntu for everyone
```

- public ipv4 change on each start-stop of instance
- when you open identity file linuxfordevops.pem then
    - in editor

linuxfordevops.pem

Private RSA Key
Strength: 2048 bits

Details

Algorithm: RSA
Size: 2048

Fingerprints

SHA1: B5 BD 5D 5D 77 3B 1F 63 B0 7F DB 2C 77 86 0B 6A E5 0C 8A 3A
SHA256: 72 43 DA D8 D3 53 E0 5A 1A 65 74 09 05 5C B8 4A 3F BA A1 4C 1A 24 67 41 CD 5E C0 DA F4 FA 93 D6

- in terminal by cat command then you can see actual private key

```
vivekant@ubuntu:~/linuxPractice$ cat linuxfordevops.pem
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAmx7/XegXjiQzXDlFVFThYXNP0sRbFTplhzI/aFGqewrYbtIJ
OOG/hIufnCURXgh/Mj6lQqZM9TG7JNgx/4JWiNVcSYqBBg+lT4JlqRG3OOhgpp6B
MY8VeL5+43BPBcmmgxiN48J7Fp/jo4Z9fn2o5APXT44zbTkcjqoSg2JJ+JeP1H5T
cDbSEh5vPl9nxWYtaw4J9wj4/nK9N7SPyTYTdRtGZvLiDgshq9AeKxg9TMYRFR7o
LUncAhw7wFF0XkbWWBNa1LPTK7eyo+beQGqIdyBb076QEe3pJwa9cmtpil7PjPLL
PqoIbNJRkJBdTdheO45c3FJCqBcYSu6dg1HxuwIDAQABAoIBAEDW6iNUmlP96cII
5mPjvvm++4S8Mt5itIGYjUyLplkcJW/dN6GhKy66NhNMoz/NCW31JZsAbHapPPKf
s/Z7grRbiuHqlRf1fkIH3XqYPx/4cj3zT0lp56264ssq7L0clw+BOf4UEuNYHg0C
cH5pBCmbOWq1o+Vp0/gGOwrUENtpyDEPHoSOTrXCMHK9kjKGRgrJbKlVe8T93HnS
bTtz2adnDSQuVMv9bWGTxvZ1I3my4MThYOdpGvzRIfhQbDBE5NOewe2XUeLoIFb8
/YE/03N8JhIJmWHEvV4oeOIPvMDr9Nxcgm7umPyo/RXYAo+fcatw1Ra6MNOXtBgf
aN+qbpECgYEAzlcx1MX/PIj4oLlNoRfM9YBgF1YkyEIJUt+GeqLO03Se7J462C3B
utztzrLg4co6Eq5pjbrx+EqSqSFzgL5u/sXWV+j1mURGuYJTYVW3WZd1N+Lvps24
F6pHB02raIUQbqAqXM5FDxHI0te+I5wRLwvgNECdGOn2C/9LXYEu3E0CgYEAwHQg
22iprRW1XUkl/XoaQByf72zk0Majhz4UfGCTbGElHkzdHGBeZYcfFgN+rhNWMuxJ
O3z/JQV0sDsmpuUBaubHMUT1YsvRKPyFCSp+9MDofGGM/wccWgLQuoMXrBYQjaqW
ii44rdEQ6tsWvW00HhhZp7l86N72fknsPqcQ6icCgYEAzMKUbawqE91stLoOQh1H
RBynvqC6kYuOixHL9qjXi9k/2FlQ1bPaet/AkYzLQuUWafFwR7hLZHREQ9gxMYJI
zURv5v8DsZuif3bH3WBdOWfmZ5Yo8aNV2w4bvDCkN21MmCsSvtPmjgS9ABYd1mKK
XuBvFgtEmjOZ+d/W3yxR30UCgYBcK4HM8uxIS9xHaA/8vKyzxcHHLhTL9EY/MDkp
dCtGvaPERjWjMHglEbiOl2/lOLeQvd840V10wo8nE70oMwchxpLhgtAu18Q9Z+ri
/+o7l+QAJcwcG7C1XyHOIZiFvy+AWZGtbgORI06B3Bmg8RsDys+tZ0VuVdIhm4+8
VdDhAwKBgQDLx5z1vRmcaa06XHb+ZV3XBB8u3ek6MZPttvIJ05qP7VJHRxuJ5/n/
5uFrc+zL0PwrmLwZdxf6GOTEty4kffpblsF9g2nDaLlfwRMGPlqHqi89Id/tH0i5
Klk0XhOv8QyPpp0LRLxtorj+0RIcMLgxSpVf9ytkPot+viZtaAQDRQ==
-----END RSA PRIVATE KEY-----vivekant@ubuntu:~/linuxPractice$
```

- for connection my laptop to vivedhara(aws) i should have "private key" and vivedhara should have "public key"
- after login by runnign above command you will see follwoing output
- to logout from ssh connection just type exit and enter you will get logout and following output you will see

- even if you close terminal ssh connection will get break

```
vivekant@ubuntu:~/linuxPractice$ ssh -i linuxfordevops.pem ubuntu@ec2-13-127-231-161.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Jan 22 09:21:57 UTC 2026

  System load:  0.0               Processes:             108
  Usage of /:   29.0% of 6.71GB   Users logged in:       0
  Memory usage: 20%               IPv4 address for enX0: 172.31.47.102
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Jan 22 09:19:26 2026 from 103.21.126.14
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-47-102:~$ exit
logout
Connection to ec2-13-127-231-161.ap-south-1.compute.amazonaws.com closed.
vivekant@ubuntu:~/linuxPractice$
```

xdg-open file_name

- it will open file in respective app.

ps

- ps command current terminal(TTY) mein chal rahe process ko dikhata hai

fuser director/file/port , fuser -v directory/file/port

`fuser`

- File / directory / port ko **kaun-sa process use** kar raha hai, batata hai
- **Sirf PID** show karta hai
- Quick check ke liye use hota hai

`fuser -v`

- File / directory / port ka **detailed process info** deta hai
- **PID + USER + COMMAND + ACCESS** dikhata hai
- Debug / permission issue samajhne me useful 👍

kill PID

- it will kill the process whose P_id is PID

free , free -h

`free`

- System ki **RAM usage** dikhata hai
- Output **KB (kilobytes)** me hota hai
- Memory debug ke liye use hota hai

`free -h`

- Same RAM info, but **human-readable format** me
- **MB / GB** me values dikhata hai
- Read karna easy hota hai 👍

- virtual memory = RAM + page file (disk)

vmstat

```
vivekant@ubuntu:~/linuxPractice$ vmstat
procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
10  0      0 1046588  43696 1586212    0    0   514    52 4980   13 18 17 64  0  0  0
vivekant@ubuntu:~/linuxPractice$
```

`vmstat` **ka output (one-shot)**

- **r = 10** → 10 processes run ke liye ready (CPU load thoda high)
- **free = 43,696 KB** → Free RAM **bahut kam**
- **buff = 1,586,212 KB / cache = 0** → RAM zyada tar buffers me use ho rahi
- **si / so = 0 / 0** → Swap **use nahi ho raha** (good sign)
- **bi / bo = 514 / 52** → Disk se thoda data read/write
- **us = 18% / sy = 17% / id = 64%** → CPU mostly **idle**
- **wa = 0%** → Disk wait nahi hai

🔁 Preemption (CPU level)

- **CPU management** ke liye hota hai
- High-priority process aate hi **CPU control cheen leta hai**
- Pehla process **pause hota hai**, terminate nahi
- Process **RAM me hi rehta hai**
- Fast hota hai (sirf CPU scheduling)

💽 Swap (RAM level)

- **Memory deficiency (RAM kam padna)** manage karne ke liye hota hai
- Process ko **RAM se disk (swap space)** me shift kiya jata hai
- Process ka **poora state disk pe save** hota hai
- Ye **memory management** ka part hai
- Slow hota hai (disk use hoti hai)

vmstat -a

- active , inactive bhi bata deta hai

```
vivekant@ubuntu:~/linuxPractice$ vmstat -a
procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
 r  b   swpd   free  inact active   si   so    bi    bo   in   cs us sy id wa st gu
 3  0      0 1023112 544820 2102948    0    0    93    11 2020    8 10  9 81  0  0  0
vivekant@ubuntu:~/linuxPractice$
```

system level command

uname , uname -a

- it will be os identity name like for mine linux for macos it give darwin.

uptime

- Load average batata hai: 👉 CPU ke liye kitne processes wait / run kar rahe hain

```
vivekant@ubuntu:~/linuxPractice$ uptime
 18:42:05 up  3:40,  1 user,  load average: 3.83, 2.50, 1.47
vivekant@ubuntu:~/linuxPractice$ uptime
 18:44:02 up  3:42,  1 user,  load average: 2.76, 2.56, 1.62
vivekant@ubuntu:~/linuxPractice$ uptime
 18:45:34 up  3:44,  1 user,  load average: 2.90, 2.60, 1.72
vivekant@ubuntu:~/linuxPractice$
```

OS har thodi der me count karta hai:

- kitne processes **running + runnable (ready)** state me hain

Phir unka **time-weighted exponential average** nikalta hai:

- last **1 min**
- last **5 min**
- last **15 min**

- Load average = kitne kaam CPU ke liye line me khade hain (1min, 5min, 15min)
- so higher value means bad
- let suppose N core cpu your pc have
- load average value=lv

```
lv>N--> overload(not good)
lv<N--> underload (good)
lv=N--> perfectely used(good)
```

man command

- it will give you details of command (what command do and their extended method)
- like "man ls" , " man clear"

lscpu

- give you all information about cpu

whoami

- it shows 👉 the user under which this terminal process is running
- it always return one user
- it give only username

who

- 👉 who shows only users who are currently logged in

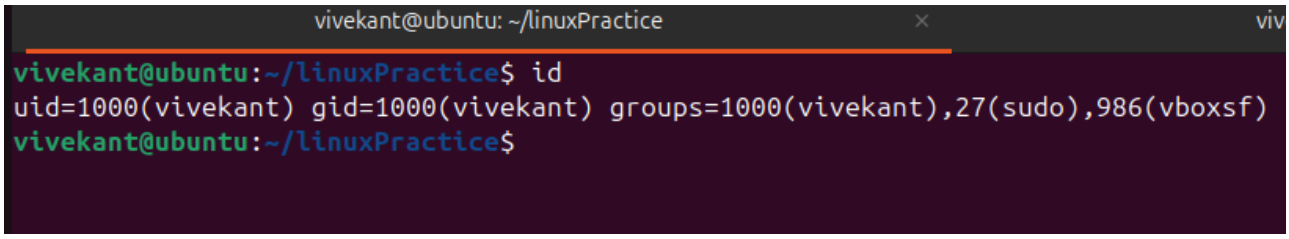
which python

- it will give you where the program is located, if not installed then it wont return anything

all command we use are file store in pc which were called on running the command so that are program actaully

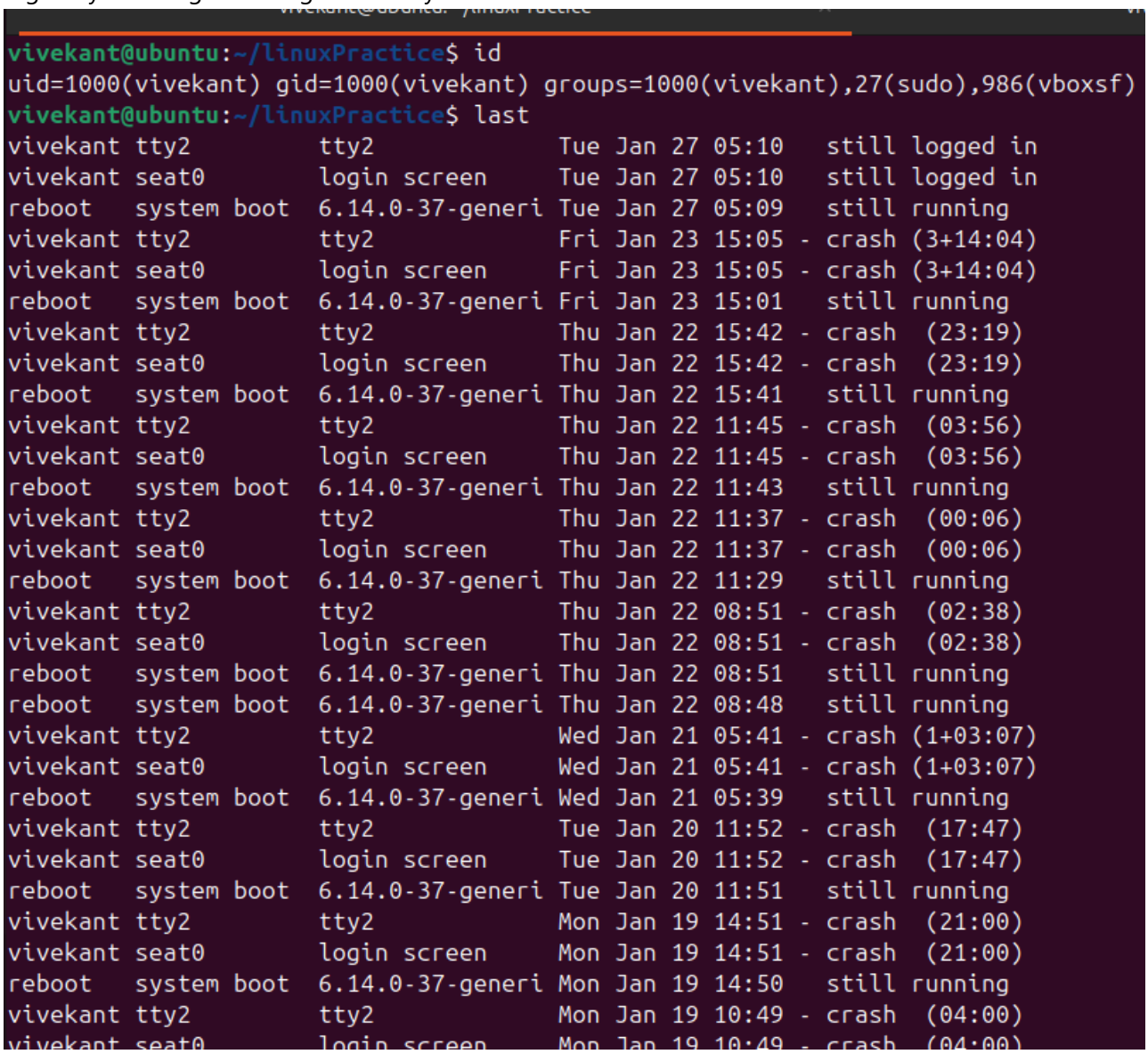
id

- its is used to check which groups and users in our system
- it give you id of all users and group inside the system
- uid->user id, gid--> group id
- only one uid mean one user only
- you can see here sudo is a group thats which they have gid



last

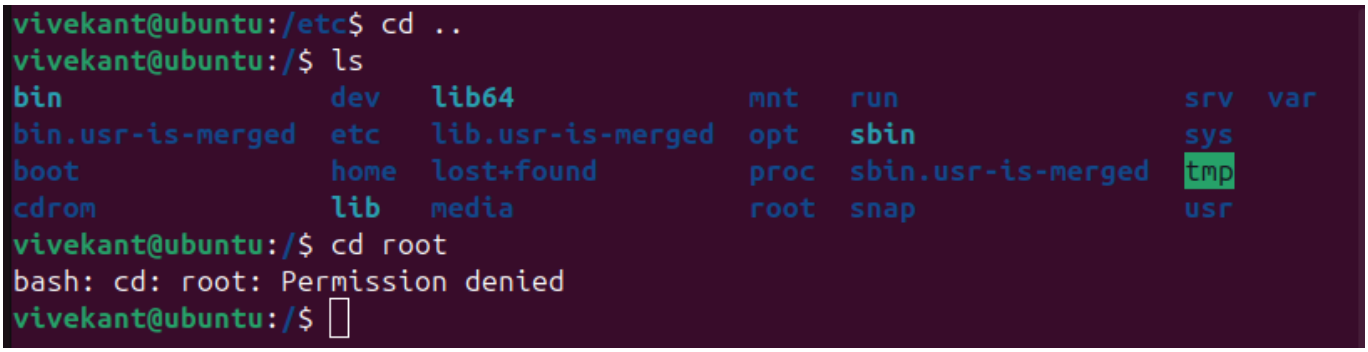
- it gives you all logIn and logOut history



cat /etc/passwd

- system ke saare users ki list + basic account info dikhata hai

you cant go inside root folder without sudo(super user do)



sudo

- its not a program, its a group which have root user
- there are lots of thing that normal user cant do , that can be done by only root user permission, for that permission we use prefix sudo command
- sudo shutdown

sudo shutdown

- turn off the system
- by thing you can stop ssh connected aws ec2 instance without going to their website

sudo reboot

- it will restart machine
- if ssh connection then you can connect again same public dns ip in aws case, i am telling this becasue every shutdown-start public dns changes of ec2 instance of aws

apt

- its application package manager user
- software download karta hai
- install karta hai
- update/remove karta hai
- Linux me automatically dependency bhi handle karta hai

apt

- apt command ke through package install, remove aur update ka short tool milta hai

apt install package

- package install karta hai but root permission ke bina kaam nahi karta

sudo apt install package

- root permission ke saath package install karta hai (recommended modern way)

sudo apt-get install package

- old/traditional method package install karne ka (abhi bhi kaam karta hai)

sudo apt remove package

- installed package ko system se uninstall/remove karta hai

which package

- if installed then it will give path otherwise nothing

User and group management command

sudo useradd -m username

- use to add user in system
- if you use -m then folder will also get created in home directory irrespective you pwd
- if you dont user -m then only user get created not folder in home directory
- when you add a new user with username then bydefault a group also created with same username and id
- when you add user in system , it only have basic permission so to give special permission root user can add this user in special group or directely give permission

```
ubuntu@ip-172-31-47-102:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-47-102:~$ sudo useradd dhara
ubuntu@ip-172-31-47-102:~$ sudo useradd -m vivekant
ubuntu@ip-172-31-47-102:~$ ls ../
ubuntu  vivekant
ubuntu@ip-172-31-47-102:~$
```

sudo passwd username

- use to create password for user in the system

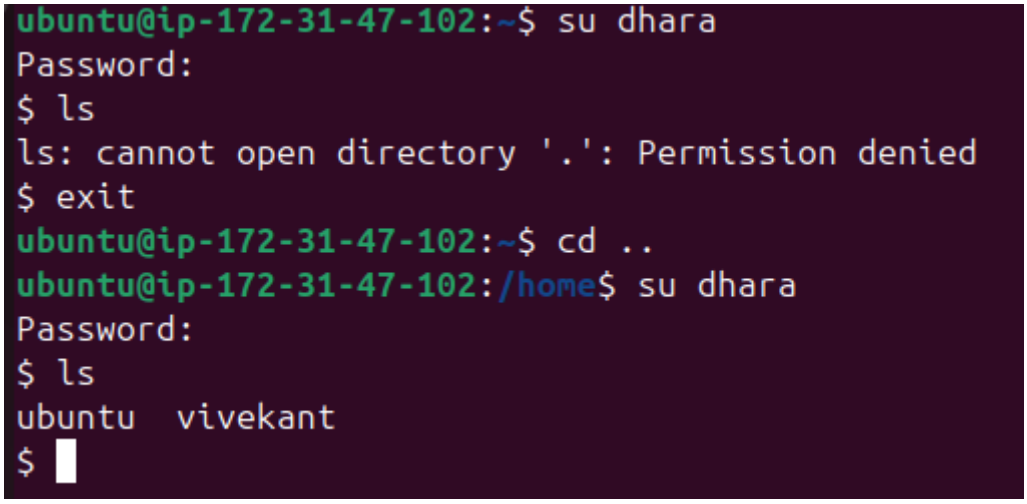
```
ubuntu@ip-172-31-47-102:~$ sudo passwd dhara
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-47-102:~$
```

su username then password

- you will switch from current user to username
- reason for error in first case is
- koi aur user dusre user ke andar file/folder bina permission ke nhi open kar sakta so, su se user change ho gaya but pwd same hai thats why error came

Ab user change ho gaya 👉 `dhara`

BUT location same rahi 👉 `/home/ubuntu`

- 

sudo userdel username

- it only delete user not directory that created while adding user

sudo userdel -r username

- delete both user and directory created while adding user.

sudo groupadd name

- it will create a group of users with name "name" in which you can add users and give permissions to all user in single time

sudo gpasswd -a username groupname

- it will add user="username" in group="groupname"
- user should pre-exist in system to add the user in the group

sudo gpasswd -M username1,username2,username3 groupname

- by this command you can add multiple user in single command

sudo groupdel groupname

- it will delete group="groupname"
- group delete karne se uske andar ka user delete nhi hota bas jo special permission mila tha users ko us group ke wajah se vo remove ho jayega

sudo gpasswd -d username groupname

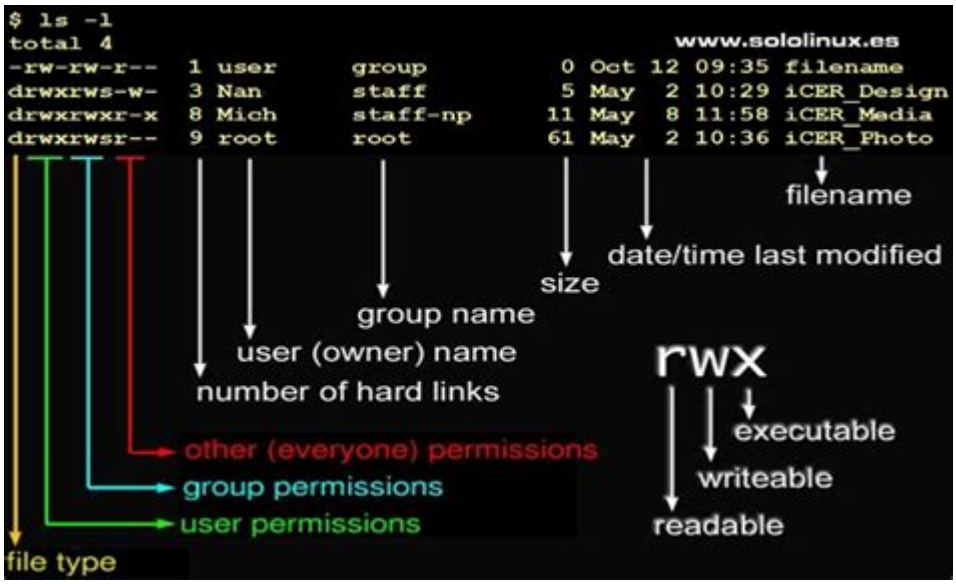
- it is used to remove user="username" from group="groupname" not from system
- removed user ke pass se vo sare permission khatam ho jayega jo us group ke vajah se mila tha

File permission commands

umask

- umask = new file/folder ki default security setting

umask value formate

- its value in order like [user][group][other] , for each one digit assign from 0-7 like following
- if some file have permission like 234 it means user have permission=2 , group have permission=3 , other have permission=4
- now permission value means---> so change that value into binary of 3 digit, let suppose permission value=3 then its binary=011
- now compare this binary with rwx--> so r=0 w=1 x=1
- 0 means no and 1 means yes
- r for read permission , w for write permission and x for execute permission
- so in above example group have permission=3 it implies read permission=0 there for the group cant read the file/folder
- so for above example file permission=234---> [010][011][100]----> [r=0 w=1 x=0][r=0 w=1 x=1][r=1 w=0 x=0]---> [-w-][-wx][r--]---> so file permission will look like -w--wxr--
- if file then --w--wxr--
- if folder then d-w--wxr--
- 

Binary	Octal	Permission	Description
000	0	---	No access
001	1	--x	Execute only
010	2	-w-	Write only
011	3	-wx	Write and Execute
100	4	r--	Read only
101	5	r-x	Read and Execute
110	6	rw-	Read and Write
111	7	rwx	Read, Write and Execute

chmod 345 file/folder

- use to change file/folder permission

```
vivekant@ubuntu:~/linuxPractice$ ls -l
total 124
-rw-rw-r-- 1 vivekant vivekant   134 Jan 28 10:41 class.txt
-rw-rw-r-- 1 vivekant vivekant 89856 Jan 22 12:08 dhara.jpg
drwxrwxr-x 3 vivekant vivekant  4096 Jan 17 12:27 dss
-rw-rw-r-- 1 vivekant vivekant    27 Jan 28 10:41 dss.txt
-r-------- 1 vivekant vivekant  1678 Jan 21 14:22 linuxfordevops.pem
lrwxrwxrwx 1 vivekant vivekant    12 Jan 17 12:32 softlink_of_vfolder -> dss/vfolder/
drwxrwxr-x 2 vivekant vivekant  4096 Jan 19 04:41 vdss
-rw-rw-r-- 1 vivekant vivekant    22 Jan 19 11:16 vdss.txt
-rw-rw-r-- 1 vivekant vivekant    24 Jan 17 20:28 vivedhara.txt
drwxrwxr-x 2 vivekant vivekant  4096 Jan 16 06:16 vivekant
-rw-rw-r-- 1 vivekant vivekant    72 Jan 15 05:48 vivekant.txt
vivekant@ubuntu:~/linuxPractice$ chmod 777 class.txt
vivekant@ubuntu:~/linuxPractice$ ls -l
total 124
-rwxrwxrwx 1 vivekant vivekant   134 Jan 28 10:41 class.txt
-rw-rw-r-- 1 vivekant vivekant 89856 Jan 22 12:08 dhara.jpg
drwxrwxr-x 3 vivekant vivekant  4096 Jan 17 12:27 dss
-rw-rw-r-- 1 vivekant vivekant    27 Jan 28 10:41 dss.txt
-r-------- 1 vivekant vivekant  1678 Jan 21 14:22 linuxfordevops.pem
lrwxrwxrwx 1 vivekant vivekant    12 Jan 17 12:32 softlink_of_vfolder -> dss/vfolder/
drwxrwxr-x 2 vivekant vivekant  4096 Jan 19 04:41 vdss
-rw-rw-r-- 1 vivekant vivekant    22 Jan 19 11:16 vdss.txt
-rw-rw-r-- 1 vivekant vivekant    24 Jan 17 20:28 vivedhara.txt
drwxrwxr-x 2 vivekant vivekant  4096 Jan 16 06:16 vivekant
-rw-rw-r-- 1 vivekant vivekant    72 Jan 15 05:48 vivekant.txt
vivekant@ubuntu:~/linuxPractice$
```

sudo chown username file/folder

- password required in this operation
- this commadn is use to change user of file/folder

```
vivekant@ubuntu:~/linuxPractice$ chmod 777 class.txt
vivekant@ubuntu:~/linuxPractice$ ls -l
total 124
-rwxrwxrwx 1 vivekant vivekant   134 Jan 28 10:41 class.txt
-rw-rw-r-- 1 vivekant vivekant 89856 Jan 22 12:08 dhara.jpg
drwxrwxr-x 3 vivekant vivekant  4096 Jan 17 12:27 dss
-rw-rw-r-- 1 vivekant vivekant    27 Jan 28 10:41 dss.txt
-r-------- 1 vivekant vivekant  1678 Jan 21 14:22 linuxfordevops.pem
lrwxrwxrwx 1 vivekant vivekant    12 Jan 17 12:32 softlink_of_vfolder -> dss/vfolder/
drwxrwxr-x 2 vivekant vivekant  4096 Jan 19 04:41 vdss
-rw-rw-r-- 1 vivekant vivekant    22 Jan 19 11:16 vdss.txt
-rw-rw-r-- 1 vivekant vivekant    24 Jan 17 20:28 vivedhara.txt
drwxrwxr-x 2 vivekant vivekant  4096 Jan 16 06:16 vivekant
-rw-rw-r-- 1 vivekant vivekant    72 Jan 15 05:48 vivekant.txt
vivekant@ubuntu:~/linuxPractice$ chown class.txt dhara
chown: invalid user: ‘class.txt’
vivekant@ubuntu:~/linuxPractice$ chown dhara class.txt
chown: changing ownership of 'class.txt': Operation not permitted
vivekant@ubuntu:~/linuxPractice$ sudo chown dhara class.txt
[sudo] password for vivekant:
vivekant@ubuntu:~/linuxPractice$ ls -l
total 124
-rwxrwxrwx 1 dhara   vivekant   134 Jan 28 10:41 class.txt
-rw-rw-r-- 1 vivekant vivekant 89856 Jan 22 12:08 dhara.jpg
drwxrwxr-x 3 vivekant vivekant  4096 Jan 17 12:27 dss
-rw-rw-r-- 1 vivekant vivekant    27 Jan 28 10:41 dss.txt
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lrwxrwxrwx 1 vivekant vivekant    12 Jan 17 12:32 softlink_of_vfolder -> dss/vfolder/
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-rw-rw-r-- 1 vivekant vivekant    24 Jan 17 20:28 vivedhara.txt
drwxrwxr-x 2 vivekant vivekant  4096 Jan 16 06:16 vivekant
-rw-rw-r-- 1 vivekant vivekant    72 Jan 15 05:48 vivekant.txt
vivekant@ubuntu:~/linuxPractice$
```

sudo chgrp username file/folder

- password required in this operation
- this command user to change group of file/folder

```
vivekant@ubuntu:~/linuxPractice$ ls -l
total 124
-rwxrwxrwx 1 dhara    vivekant   134 Jan 28 10:41 class.txt
-rw-rw-r-- 1 vivekant vivekant 89856 Jan 22 12:08 dhara.jpg
drwxrwxr-x 3 vivekant vivekant  4096 Jan 17 12:27 dss
-rw-rw-r-- 1 vivekant vivekant    27 Jan 28 10:41 dss.txt
-r-------- 1 vivekant vivekant  1678 Jan 21 14:22 linuxfordevops.pem
lrwxrwxrwx 1 vivekant vivekant    12 Jan 17 12:32 softlink_of_vfolder -> dss/vfolder/
drwxrwxr-x 2 vivekant vivekant  4096 Jan 19 04:41 vdss
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-rw-rw-r-- 1 vivekant vivekant    24 Jan 17 20:28 vivedhara.txt
drwxrwxr-x 2 vivekant vivekant  4096 Jan 16 06:16 vivekant
-rw-rw-r-- 1 vivekant vivekant    72 Jan 15 05:48 vivekant.txt
vivekant@ubuntu:~/linuxPractice$ sudo chgrp dhara class.txt
vivekant@ubuntu:~/linuxPractice$ ls -l
total 124
-rwxrwxrwx 1 dhara    dhara      134 Jan 28 10:41 class.txt
-rw-rw-r-- 1 vivekant vivekant 89856 Jan 22 12:08 dhara.jpg
drwxrwxr-x 3 vivekant vivekant  4096 Jan 17 12:27 dss
-rw-rw-r-- 1 vivekant vivekant    27 Jan 28 10:41 dss.txt
-r-------- 1 vivekant vivekant  1678 Jan 21 14:22 linuxfordevops.pem
lrwxrwxrwx 1 vivekant vivekant    12 Jan 17 12:32 softlink_of_vfolder -> dss/vfolder/
drwxrwxr-x 2 vivekant vivekant  4096 Jan 19 04:41 vdss
-rw-rw-r-- 1 vivekant vivekant    22 Jan 19 11:16 vdss.txt
-rw-rw-r-- 1 vivekant vivekant    24 Jan 17 20:28 vivedhara.txt
drwxrwxr-x 2 vivekant vivekant  4096 Jan 16 06:16 vivekant
-rw-rw-r-- 1 vivekant vivekant    72 Jan 15 05:48 vivekant.txt
vivekant@ubuntu:~/linuxPractice$
```

compression command

zip zipped_file_name.zip file

zip -r zipped_folder.zip folder

- these command are used to zip file and folder

unzip zipped_file.zip

- this command user to unzip the file or folder

gunzip is also a another tool to zip and unzip file/folder which ends with .gz

gunzip is also another tool used to zip and unzip files/folders which end with `.gz`

tar -cvzf compressed_filename.tar.gz folder/file

- it creates an archive and compresses the folder/file using gzip
- archive matlab bahut sare file ko ek sath jor ke rakhana
- `cvzf` meaning:
  - `c` → create (naya archive banana)
  - `v` → verbose (screen par dikhata hai kya process ho raha hai)
  - `z` → gzip compression use karta hai
  - `f` → file name specify karta hai

tar -xvzf compressed_filename.tar.gz

- this is used to extract files/folders from compressed archive
- `xvzf` meaning:
  - `x` → extract (archive ko kholna)
  - `v` → verbose (details dikhata hai)
  - `z` → gzip decompress karta hai
  - `f` → file name

file transfer commands

scp -i key_pair_file.pem source_file destination

- scp --> secure copy
- it is use for send file="source_file" to server(ec2 aws) from our local computer.
- for your pc to aws

```
source_file --> path of file in you local computer
destination username@publicDNS:/home/username/
or destination username@publicDNS:/home/username/yourDesiredFileName with
extension
```

- for aws to pc

```
source_file --> path of file in aws
ex. username@publicDNS:/home/username/filepath
destination :
- . for current folder
- ./YourDesirefile name with extension
```

scp -i key_pair_file.pem -r source_file destination

- use to copy folder from one server to another server
- other thing remain same as file copy

rsync -e "ssh -i linuxfordevops.pem" -avz source destination

- -avz means archive(sare file adn folder ek sath jodna), v means verbose(visula dikhana kya sab process ho raha hai), z mean zip karke send/receive karna
- -e ka matlab ham remote shell type dene ja rahe hai like ssh , rsh
- -avz ka use nhi karoge toh folder copy nhi hoga and yadi folder hai directory ke andar toh other file change bhi sync nhi hoga so atlease -av ka use jarur kare

rsync -e "ssh -i linuxfordevops.pem" -av ubuntu@ec2-65-1-85-165.ap-south-1.compute.amazonaws.com:/home/ubuntu/vivekant/ .

- isme vivekant/ hai jiska matlab vivekant folder ko sync nhi iske andar ke content ko sync karna hai

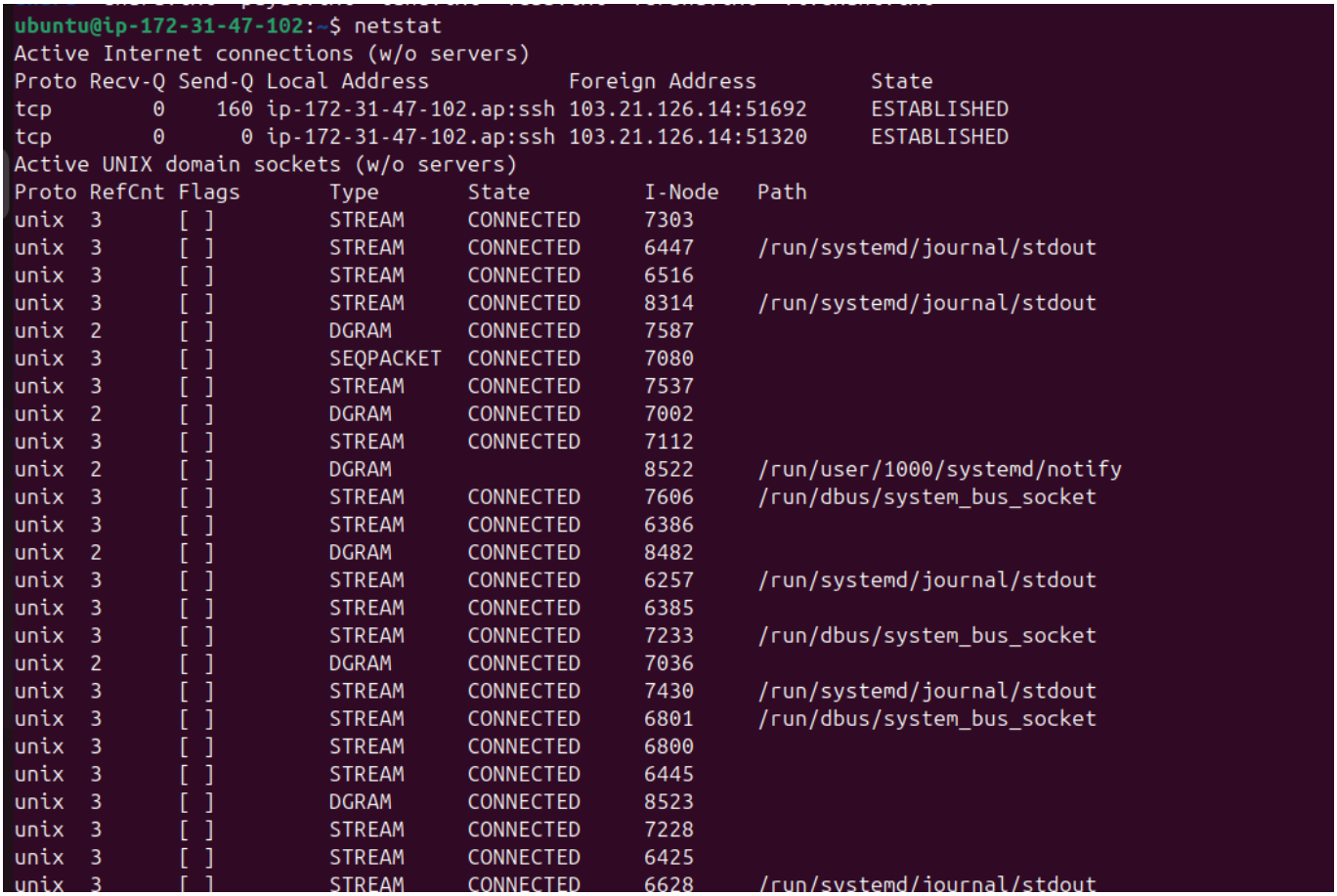
- sirf vivekant rakhoge toh vivekant folder ko sync karna hoga, then syncing mein rsync vivekant folder dhundhega in destination.

networking

ping domain

- use to check we(client and server) are able to send and receive data between client and server
- example ping google.com--> it will check i am able to send data to google.com and gooogle.com able to send data to me

netstat or ss



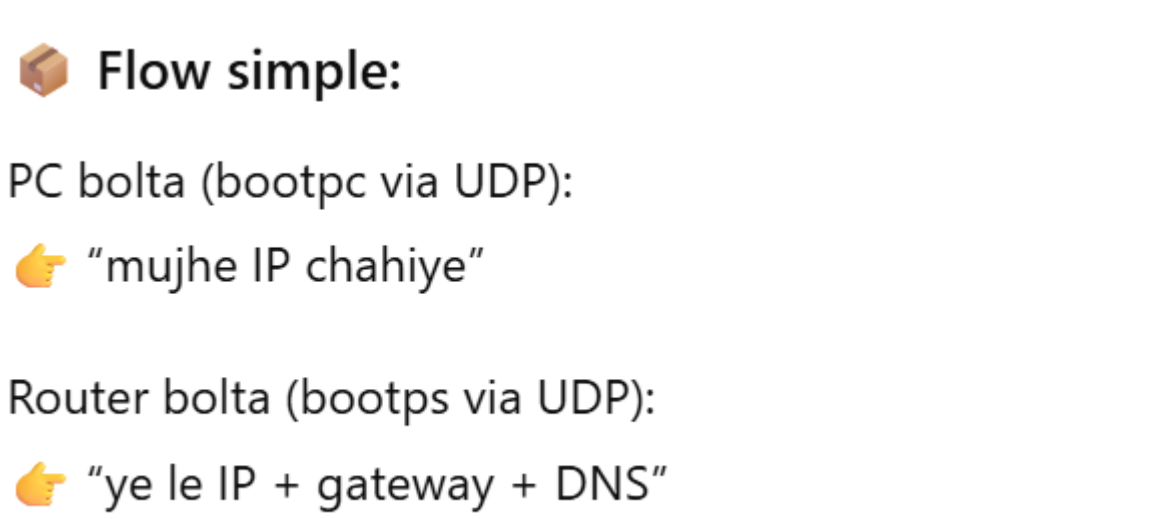
```
ubuntu@ip-172-31-47-102:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0    160 ip-172-31-47-102.ap:ssh 103.21.126.14:51692     ESTABLISHED
tcp        0      0 ip-172-31-47-102.ap:ssh 103.21.126.14:51320     ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     7303
unix  3      [ ]         STREAM     CONNECTED     6447     /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     6516
unix  3      [ ]         STREAM     CONNECTED     8314     /run/systemd/journal/stdout
unix  2      [ ]         DGRAM      CONNECTED     7587
unix  3      [ ]         SEQPACKET  CONNECTED     7080
unix  3      [ ]         STREAM     CONNECTED     7537
unix  2      [ ]         DGRAM      CONNECTED     7002
unix  3      [ ]         STREAM     CONNECTED     7112
unix  2      [ ]         DGRAM                    8522     /run/user/1000/systemd/notify
unix  3      [ ]         STREAM     CONNECTED     7606     /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     6386
unix  2      [ ]         DGRAM      CONNECTED     8482
unix  3      [ ]         STREAM     CONNECTED     6257     /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     6385
unix  3      [ ]         STREAM     CONNECTED     7233     /run/dbus/system_bus_socket
unix  2      [ ]         DGRAM      CONNECTED     7036
unix  3      [ ]         STREAM     CONNECTED     7430     /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     6801     /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     6800
unix  3      [ ]         STREAM     CONNECTED     6445
unix  3      [ ]         DGRAM      CONNECTED     8523
unix  3      [ ]         STREAM     CONNECTED     7228
unix  3      [ ]         STREAM     CONNECTED     6425
unix  3      [ ]         STREAM     CONNECTED     6628     /run/systemd/journal/stdout
```

- you can see two connection there from you ec2 of aws connected with you pc
- note one thing 2 connection established for each ssh connection
- ssh uses one channel for commands + terminals , another channel for data transfer
- baad baki jo bhi connection hai vo linux ke internal connection hai jo unix wagera dikha raha hai
- yaha jo forein address hai vo hamara pc ka ip hai
- 👉 Ye netstat output EC2 server ka hai
- 👉 Ye dikha raha hai kaun-kaun internet se EC2 se connected hai
- 👉 Tere ghar ka Wi-Fi / laptop ka network yahan matter nahi karta

```
vivekant@ubuntu:~/linuxPractice$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 ubuntu:57952            ec2-65-1-85-165.ap-:ssh ESTABLISHED
udp        0      0 ubuntu:bootpc           _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     20675    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     17914
unix  3      [ ]         STREAM     CONNECTED     17816    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     16779
unix  3      [ ]         STREAM     CONNECTED     11205    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     15792    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     11097
unix  2      [ ]         DGRAM      CONNECTED     14166
unix  3      [ ]         STREAM     CONNECTED     18918
unix  3      [ ]         STREAM     CONNECTED     14306    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     14201
unix  3      [ ]         STREAM     CONNECTED     18063    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     15099
unix  3      [ ]         STREAM     CONNECTED     13743    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     18114
unix  3      [ ]         STREAM     CONNECTED     17966    /run/user/1000/pulse/native
unix  3      [ ]         STREAM     CONNECTED     16799    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     18829    /home/vivekant/.cache/ibus/dbus-8drn1Ek0
unix  3      [ ]         STREAM     CONNECTED     11169
unix  3      [ ]         STREAM     CONNECTED     16577    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     8892
unix  3      [ ]         STREAM     CONNECTED     20689    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     20541
unix  3      [ ]         STREAM     CONNECTED     16080    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     15139
```

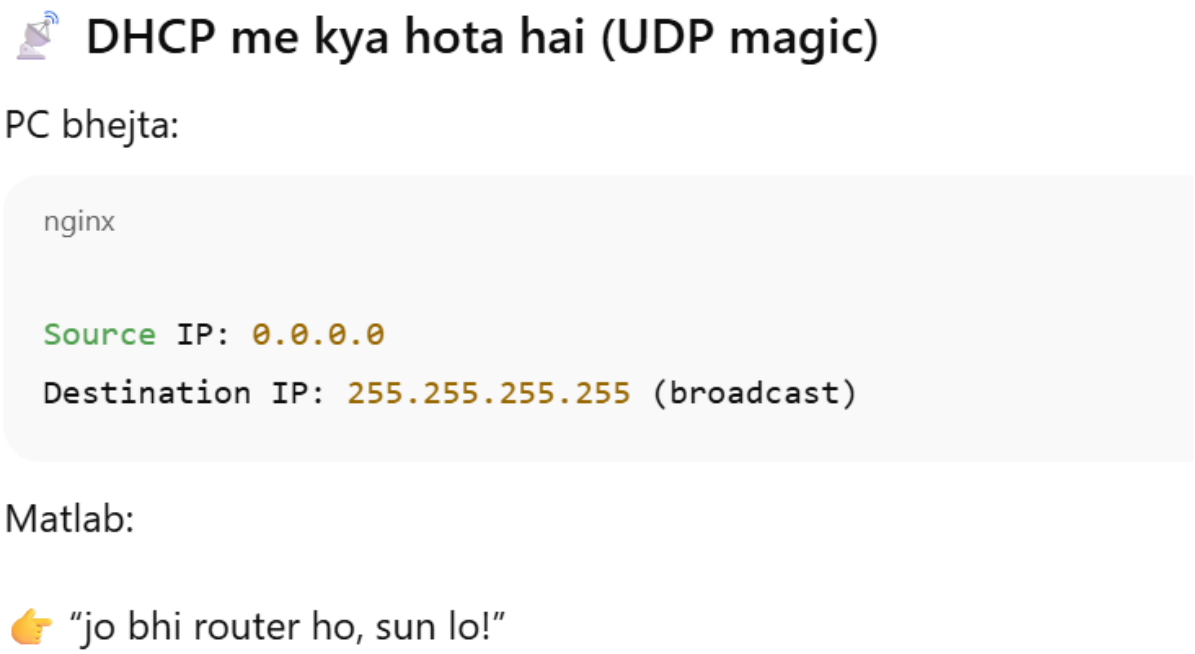
- ye connection or stat mere pc ka hai
- yaha pe do connection hai ,first mere pc ubuntu with port=57952 se aws ke ec2 via tcp protocol
- dusra connection hai mere pc ubuntu:bootpc se wifi ke bootps se via udp protocol jisse wifi internet mil raha hai

  - **bootpc** = PC jo IP maang raha hai (DHCP client)
  - **bootps** = router/modem jo IP deta hai (DHCP server)
  - **UDP** = jis raaste se ye baat-cheet hoti hai

- 

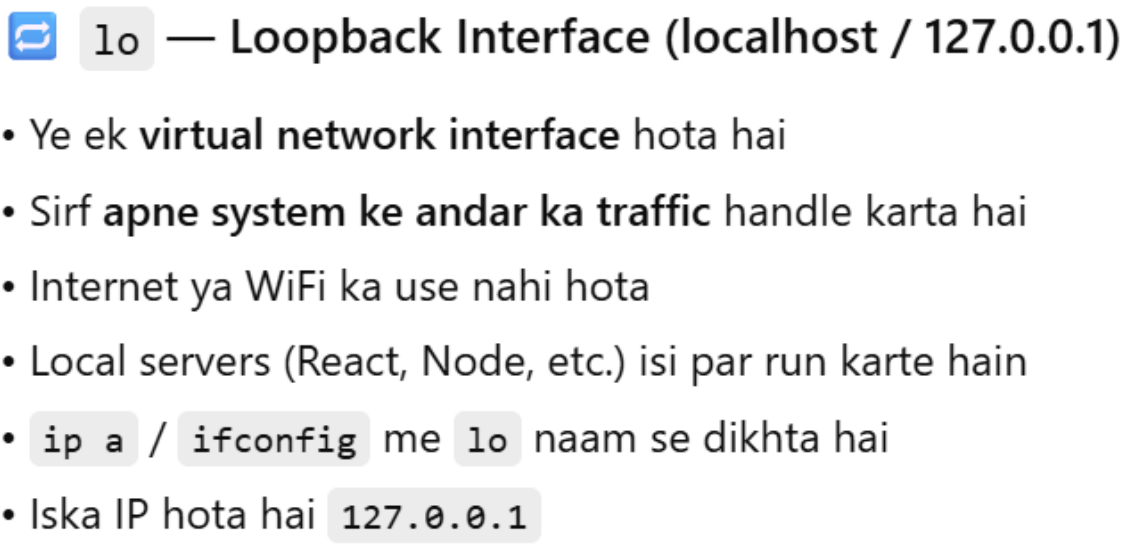
- bootpc , bootps bas role ka naam hai, koi procedure nad protocol nhi

- 👉 Broadcast ka matlab: local network me sabko bhejna



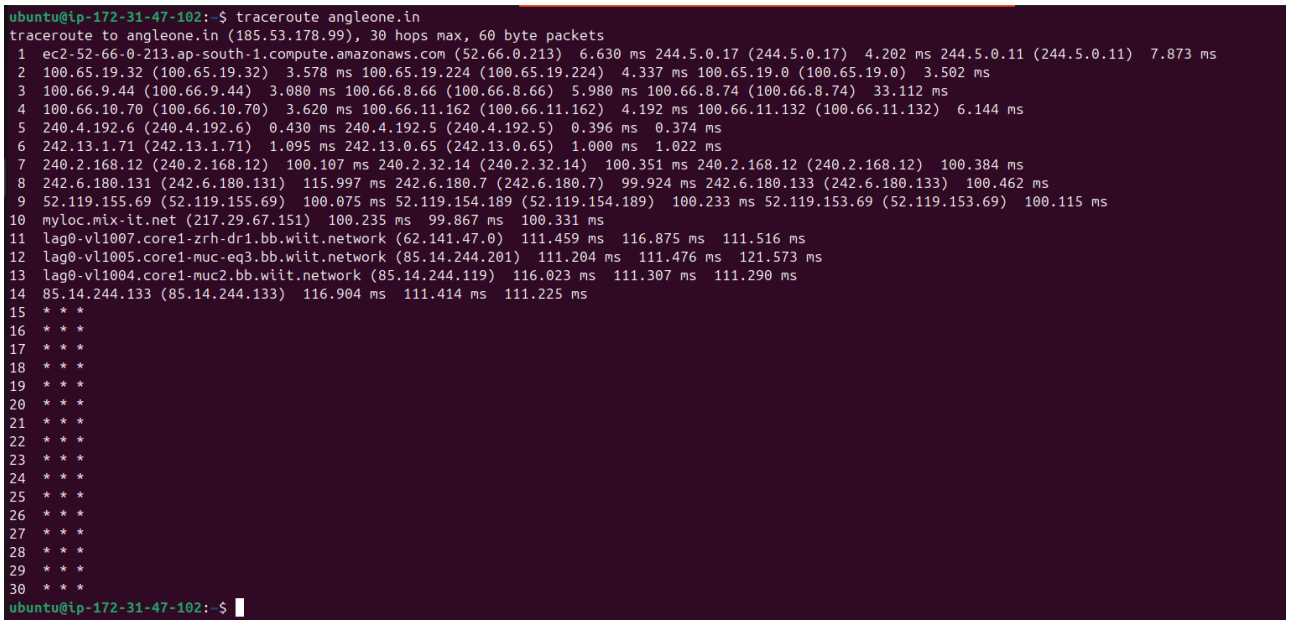
ifconfig

- inet-->private ip
- inet6---> ipv6 hota hai
- ye sare network interface ko dikhata hai ki kon kon sa network interface hai hamare system mein
- jab ham docker install karte hai toh vo apna ek network interface ban leta hai



traceroute google.com

- use to find ip hopes on hitting request on server, like bich mein kitne server se hoker request final server ke pass pahuch , us sare server ka ip ye dega



```
ubuntu@ip-172-31-47-102:~$ traceroute angleone.in
traceroute to angleone.in (185.53.178.99), 30 hops max, 60 byte packets
 1  ec2-52-66-0-213.ap-south-1.compute.amazonaws.com (52.66.0.213)  6.630 ms 244.5.0.17 (244.5.0.17)  4.202 ms 244.5.0.11 (244.5.0.11)  7.873 ms
 2  100.65.19.32 (100.65.19.32)  3.578 ms 100.65.19.224 (100.65.19.224)  4.337 ms 100.65.19.0 (100.65.19.0)  3.502 ms
 3  100.66.9.44 (100.66.9.44)  3.080 ms 100.66.8.66 (100.66.8.66)  5.980 ms 100.66.8.74 (100.66.8.74)  33.112 ms
 4  100.66.10.70 (100.66.10.70)  3.620 ms 100.66.11.162 (100.66.11.162)  4.192 ms 100.66.11.132 (100.66.11.132)  6.144 ms
 5  240.4.192.6 (240.4.192.6)  0.430 ms 240.4.192.5 (240.4.192.5)  0.396 ms  0.374 ms
 6  242.13.1.71 (242.13.1.71)  1.095 ms 242.13.0.65 (242.13.0.65)  1.000 ms  1.022 ms
 7  240.2.168.12 (240.2.168.12)  100.107 ms 240.2.32.14 (240.2.32.14)  100.351 ms 240.2.168.12 (240.2.168.12)  100.384 ms
 8  242.6.180.131 (242.6.180.131)  115.997 ms 242.6.180.7 (242.6.180.7)  99.924 ms 242.6.180.133 (242.6.180.133)  100.462 ms
 9  52.119.155.69 (52.119.155.69)  100.075 ms 52.119.154.189 (52.119.154.189)  100.233 ms 52.119.153.69 (52.119.153.69)  100.115 ms
10  myloc.mix-it.net (217.29.67.151)  100.235 ms  99.867 ms  100.331 ms
11  lag0-vl1007.core1-zrh-dr1.bb.wiit.network (62.141.47.0)  111.459 ms  116.875 ms  111.516 ms
12  lag0-vl1005.core1-muc-eq3.bb.wiit.network (85.14.244.201)  111.204 ms  111.476 ms  121.573 ms
13  lag0-vl1004.core1-muc2.bb.wiit.network (85.14.244.119)  116.023 ms  111.307 ms  111.290 ms
14  85.14.244.133 (85.14.244.133)  116.904 ms  111.414 ms  111.225 ms
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
ubuntu@ip-172-31-47-102:~$
```

🌐 Traceroute Output — `* * *` & Reserved IPs

⭐ `* * *` ka matlab:

- Router ne reply nahi diya
- ICMP block/firewall laga hota hai
- Security ke liye hops hide kiye jaate hain
- Ye normal hai (error nahi)

🔒 Reserved / Private IP kyu dikhte hain:

- Ye public internet ke IP nahi hote
- ISP aur AWS jaise networks internally use karte hain
- Location trace nahi hoti
- Backbone routing ke liye hote hain

Examples:

`10.x.x.x`

`100.x.x.x`

`172.16-31.x.x`

`240.x.x.x`

🎯 Summary:

- `* * *` = hop silent
- Reserved IP = private network path
- Traceroute normally aisa hi dikhta hai

🧠 One-line memory:

> Stars = no reply
> Reserved IP = internal internet roads

tracepath google.com

- ye batata hai ki Source se destination tak network ka route + delay + MTU issues detect karna (without root permission)

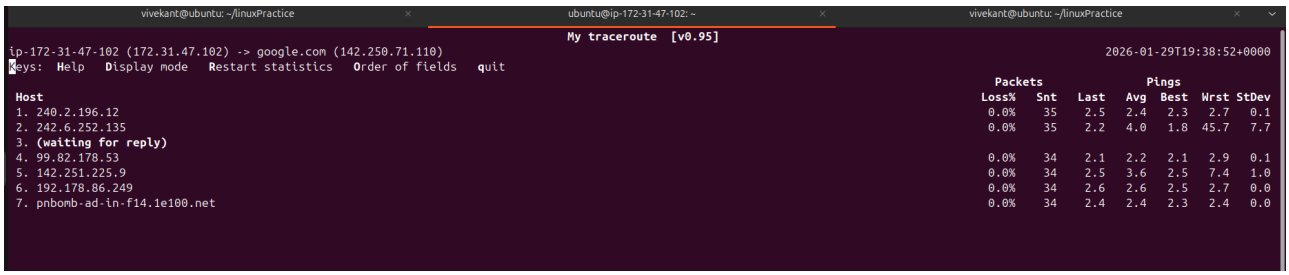
mtr google.com

🔧 DevOps me `mtr` ka use:

- Network latency find karna (kaunsa hop slow hai)
- Packet loss kaha ho raha hai check karna
- Server reachability debug karna
- ISP vs cloud issue identify karna
- Production outage me route problem pakadna

📊 Why mtr powerful hai:

- Ping + traceroute dono ek saath
- Live updating stats
- Har hop ka loss & delay dikhta hai

- 

nslookup domain

telnet domain

- ye hame batata hai ki request receiving server use which port?

🌐 DevOps me `nslookup` ka use:

- Domain ka IP address find karna
- DNS working hai ya nahi check karna
- Kaunsa DNS server response de raha hai dekhna
- Website down ho to DNS issue isolate karna
- Load balancer ka IP verify karna

🔌 DevOps me `telnet` ka use:

- Server ka port open hai ya nahi check karna
- App service running hai ya nahi verify karna
- Firewall block kar raha hai ya nahi test karna
- DB / API connectivity quick test
- Production connectivity debugging

🧠 One-line memory:

> nslookup = DNS checker
> telnet = port connectivity tester

🌐 `Server: 127.0.0.53`

👉 Ye tera **local DNS resolver** hai
(systemd-resolved in Ubuntu)
👉 Ye aage real DNS servers se poochta hai

📡 `Address: 127.0.0.53#53`

👉 DNS server ka IP + port
👉 **53 = DNS ka standard port**

📄 `Non-authoritative answer:`

👉 Reply direct google ke DNS se nahi
👉 Cache ya forwarder ke through aaya
👉 Normal cheez hai (error nahi)

🌍 `Name: google.com`

👉 Domain jo tune query kiya

📍 `Address: 142.250.207.142`

👉 Google ka **IPv4 address**

📍 `Address: 2404:6800:4009:825::200e`

👉 Google ka **IPv6 address**

---

differenece between nslookup and traceroute

- actually traceroute packet bhejta hai final server tak pahuchana ke liye and jo hope(ip) milta hai subko return karta hai
- nslookup DNS server se baat karke domain ka ip find karta hai

- first ip is ipv4 and second one is ipv6

```
vivekant@ubuntu:~/linuxPractice$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.206.174
Name:   google.com
Address: 2404:6800:4009:824::200e

vivekant@ubuntu:~/linuxPractice$
```

telnet domain port_number

- telnet ek network protocol hai jo remote server se interact karne ke liye use hota hai. Iski madad se hum door ki machines se connect hoke commands chala sakte hain aur unke saath directly communication kar sakte hain. Ye mainly network debugging, port connection check karne aur basic server interaction ke liye use hota hai. Aaj-kal iski jagah zyada secure protocol jaise SSH ne le li hai, lekin phir bhi telnet abhi bhi kuch basic testing ke liye kaafi useful hai.
- server ka http(port=80) , https(port=443) responsive hai ya nhi ye check karne ke liye use hota hai mainly

hostname

- ip bolne mein complicated lagta hai so we assign a name to a ip of our system that is called hostname , like for 127.0.0.1 is ip we use localhost as hostname , google.com is hostname and their real ip will different which look complicated.

```
ubuntu@ip-172-31-47-102:~$ hostname
ip-172-31-47-102
ubuntu@ip-172-31-47-102:~$ cat /etc/hosts
127.0.0.1 localhost

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
ubuntu@ip-172-31-47-102:~$
```

ip address show

- ip address show command ka use humare system ke saare network interfaces ki detail dekhne ke liye hota hai. Isse hume har interface ka IP address, subnet mask, broadcast address aur uska status pata chalta hai, jo network troubleshooting aur configuration me help karta hai.

ip/if/iw + config

- internet protocol, interfaces , interface wireless,
- ipconfig new tool which replaced the ifconfig
- dono network interfaces ka details dekhane ke liye kiya jata hai
- iwconfig use for to see and change wifi interfaces

nmcli device status

- show all network interfaces

dig domain

- it will give you all the information about where the website is deployed so overall ip address you will,
- it also give you ip by asking to dns server
- so final ip of traceroute, nslookup ip and dig ip all three are same but each command have different purpose

```
vivekant@ubuntu:~/linuxPractice$ dig angleone.in

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> angleone.in
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 45728
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;angleone.in.                   IN      A

;; ANSWER SECTION:
angleone.in.            3600    IN      A       185.53.178.99

;; Query time: 124 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Jan 31 19:22:50 UTC 2026
;; MSG SIZE  rcvd: 56

vivekant@ubuntu:~/linuxPractice$
```

whois

- use to find all the information about domain, from where it bought , who is the owner etc.

arp

- it shows IP address to MAC address mapping of devices in your local network

```
vivekant@ubuntu:~/linuxPractice$ arp
Address                  HWtype  HWaddress           Flags Mask            Iface
_gateway                 ether   52:55:0a:00:02:02   C                     enp0s3
vivekant@ubuntu:~/linuxPractice$
```

Gateway ka simple matlab:

> Gateway = tumhara router / network ka exit door to internet

Jis device ke through:

PC → bahar internet → websites → servers

ifplugstatus

- it give you, which network interface are in use and active and which are not active(unplygd)

```
ubuntu@ip-172-31-47-102:~$ ifplugstatus
lo: link beat detected
enX0: link beat detected
docker0: unplugged
ubuntu@ip-172-31-47-102:~$
```

- docker0 is not running right now.

curl -X GET https://dummy-json.mock.beeceptor.com/posts

- Curl ka use hota hai kisi bhi URL se data laane ke liye — chahe wo koi API call ho, ya koi webpage. Basically, ye command-line se internet par GET, POST jaise requests bhejne aur response paane ka tool hai.
- -X ke jagah yadi aap -O rakh de toh file download kar sakte hai lekin link final file ka ho for example

curl -O https://file-examples.com/wp-content/storage/2017/02/file-sample_100kB.doc

- this is for download the file
- link final file ka ho

wget https://file-examples.com/wp-content/storage/2017/02/file-sample_100kB.doc

curl -X GET https://dummy-json.mock.beeceptor.com/posts | jq

- sometimes output came in raw formate which is not human readible, so to make that human readable and good formate we use jq command thats its the use of jq
- jq will show raw output json data into good looking data and human readible

nmap -v domain

- -v mean verbose(dikhao)
- use to scan a website
- you can see which ports are open if other port are remain open by mistake you can attack on that port and poeple do that
- for website port 80(http) , 443 (https) are generally opened

```
ubuntu@ip-172-31-47-102:~$ nmap -v angleone.in
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-01 10:52 UTC
Initiating Ping Scan at 10:52
Scanning angleone.in (185.53.178.99) [2 ports]
Completed Ping Scan at 10:52, 0.12s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 10:52
Completed Parallel DNS resolution of 1 host. at 10:52, 0.18s elapsed
Initiating Connect Scan at 10:52
Scanning angleone.in (185.53.178.99) [1000 ports]
Discovered open port 80/tcp on 185.53.178.99
Completed Connect Scan at 10:52, 9.56s elapsed (1000 total ports)
Nmap scan report for angleone.in (185.53.178.99)
Host is up (0.12s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http

Read data files from: /usr/bin/../share/nmap
Nmap done: 1 IP address (1 host up) scanned in 10.04 seconds
ubuntu@ip-172-31-47-102:~$
```

route

Linux me `route` **command ka kaam hota hai network ka rasta (path) dekhna aur control karna** — matlab aapka system data ko **kis gateway ya interface se bhej raha hai.**

Short me:

👉 **Route command batata hai data internet ya dusre network tak kis raaste se jaa raha hai**

Simple samjho:

Jaise real life me:

> Ghar se office jaane ka route hota hai

Waise hi Linux me:

> System se internet tak jaane ka route hota hai

Common uses:

📌 **Current routes dekhne ke liye:**

bash Copy code

```
route
```

ya

bash Copy code

```
ip route
```

Ye dikhata hai:
• default gateway
• kaunsa network kis interface se ja raha hai

📌 **Default gateway ka matlab:**
👉 Ye main rasta hota hai jahan se internet pe data jaata hai

One line me:

Route command = network traffic ka road map

Agar chaho to main bataun:

• route vs ip route difference
• gateway kya hota hai clearly
• real example with output explain

```
ubuntu@ip-172-31-47-102:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         ip-172-31-32-1. 0.0.0.0         UG    100    0        0 enX0
172.17.0.0      0.0.0.0         255.255.0.0     U     0      0        0 docker0
ip-172-31-0-2.a ip-172-31-32-1. 255.255.255.255 UGH   100    0        0 enX0
172.31.32.0     0.0.0.0         255.255.240.0   U     100    0        0 enX0
ip-172-31-32-1. 0.0.0.0         255.255.255.255 UH    100    0        0 enX0
ubuntu@ip-172-31-47-102:~$
```

ps aux

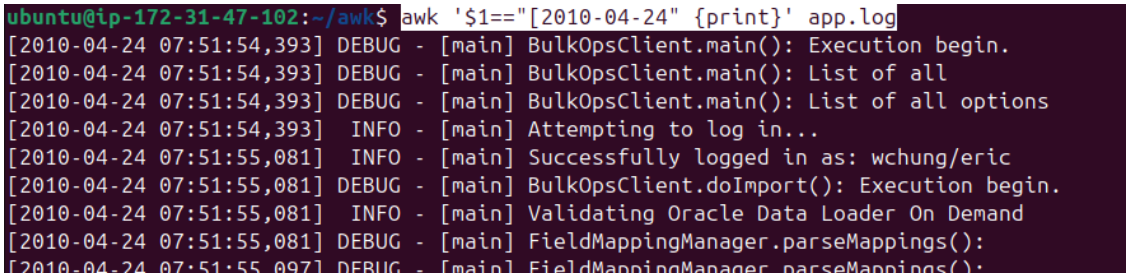
- ps aux Linux me system ke saare running processes (programs) dikhaata hai — kaun sa program chal raha hai, kis user ne start kiya, kitna CPU/RAM use ho raha hai, etc.

PRO Linux command (little bit programming)

awk

- it only work on CSV(comman seprated value) or TSV(tab/space seprated value) data
- for any kind of file , we use sed(stream editor)
- syntax for awk will be

```
awk 'script' file
```

- example
  - awk '{print}' app.log
  - awk '{print $2}' app.log
  - awk '{print $2, $3}' app.log
  - awk '{print $2,$5}' app.log
  - awk '/INFO/ {print $1, $2 , $5}' app.log
  - awk '/INFO/ {print}' app.log
    - it will print all line which contain 'INFO'
  - awk '/INFO/ {count++} END {print "The number of line which contains 'INFO' = " count}' app.log
    - it will print how many line which contains 'INFO'
  - awk '$1=="[2010-04-24" {print}' app.log
    - it will seprate all the log of date 2010-04-24
    - 

  - awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,1000]" {print}' app.log
    - it will print all log of date 2010-04-24 and time 7:51:00,000 to 7:52:00,000

```
ubuntu@ip-172-31-47-102:~/awk$ awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,1000]" {print}' app.log
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): Execution begin.
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all options
[2010-04-24 07:51:54,393]  INFO - [main] Attempting to log in...
[2010-04-24 07:51:55,081]  INFO - [main] Successfully logged in as: wchung/eric
[2010-04-24 07:51:55,081] DEBUG - [main] BulkOpsClient.doImport(): Execution begin.
[2010-04-24 07:51:55,081]  INFO - [main] Validating Oracle Data Loader On Demand
[2010-04-24 07:51:55,081] DEBUG - [main] FieldMappingManager.parseMappings():
[2010-04-24 07:51:55,097] DEBUG - [main] FieldMappingManager.parseMappings():
[2010-04-24 07:51:55,331] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run():
[2010-04-24 07:51:55,331]  INFO - [main] A SOAP request was sent to the server to
[2010-04-24 07:51:55,862] DEBUG - [Thread-1]
[2010-04-24 07:51:55,862] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run():
[2010-04-24 07:51:55,862] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run(): SOAP
[2010-04-24 07:51:55,862] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run(): Going
[2010-04-24 07:51:55,862] DEBUG - [main]
[2010-04-24 07:51:55,862] DEBUG - [main] There was an error sending the SOAP request
[2010-04-24 07:51:55,862] DEBUG - [main] BulkOpsClient.sendValidationRequest():
[2010-04-24 07:51:56,862]  INFO - [main] A SOAP request was sent to the server to
```

- awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,000]" {count++} END {print count}' app.log
  - it will give you total count of log line of desire date and time
  - millisecond shoudl be 000 to 999 if you put 1000 it will consider as hour:min:second+1,000
  -

```
ubuntu@ip-172-31-47-102:~/awk$ awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,000]" {count++} END {print count}' app.log
19
```

- awk 'NR>=2 && NR<=10 {print}' app.log
  - it will give you second to 10th lines of log
  -

```
ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10' app.log

[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all
configurations loaded: {numofthreads=1, impstatchkinterval=30, maxloginattempts=1,
manifestfiledir=.\Manifest\, sessionkeepchkinterval=300, routingurl=https://
sso.crmondemand.com, hosturl=http://sdchs20n263.us.oracle.com, testmode=debug,
maxthreadfailure=1, logintimeoutms=180000, csvblocksize=1000, maxsoapsize=10240}

[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all options
loaded: {password=*********, clientloglevel=detailed, waitforcompletion=False,
```

- awk 'NR>=2 && NR<=10 {print NR}' app.log
- it will print row number from 2 to 10
-

```
ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10 {print NR}' app.log
2
3
4
5
6
7
8
9
10
```

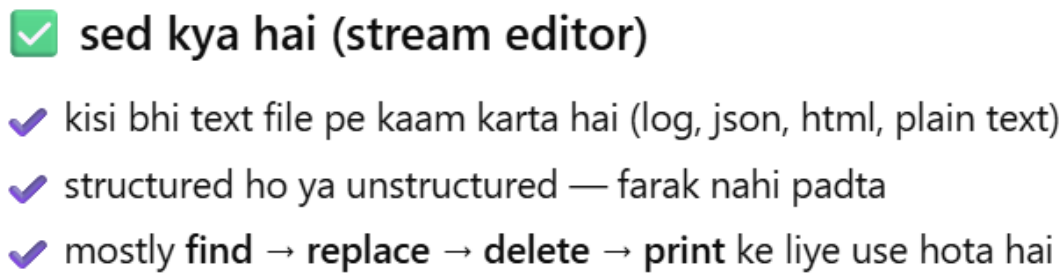
- awk 'NR>=2 && NR<=10 {print NR, $0}' app.log
  - it will pring row number with corresponding log
  - $0 mean all column or whole line of a row

```
ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10 {print NR, $0}' app.log
2
3 [2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all
4 configurations loaded: {numofthreads=1, impstatchkinterval=30, maxloginattempts=1,
5 manifestfiledir=.\Manifest\, sessionkeepchkinterval=300, routingurl=https://
6 sso.crmondemand.com, hosturl=http://sdchs20n263.us.oracle.com, testmode=debug,
7 maxthreadfailure=1, logintimeoutms=180000, csvblocksize=1000, maxsoapsize=10240}
8
9 [2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all options
10 loaded: {password=*********, clientloglevel=detailed, waitforcompletion=False,
ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10 {print NR, $0}' app.log
```

difference between awk and sed

- awk work only on structured data csv or tsv
- sed works line by line irrespective of data formate structured or unstructured
- syntax difference is there

sed



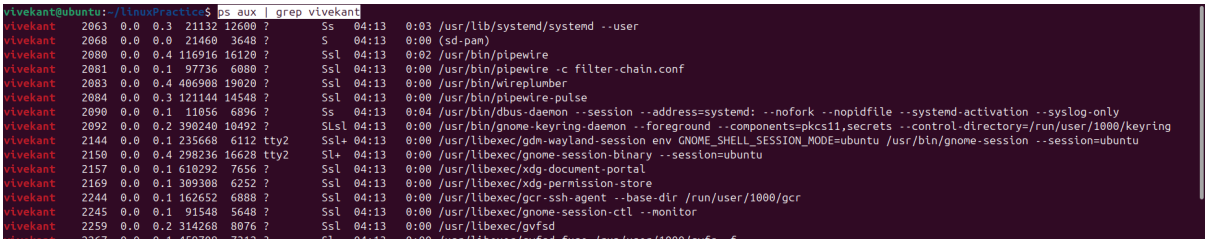
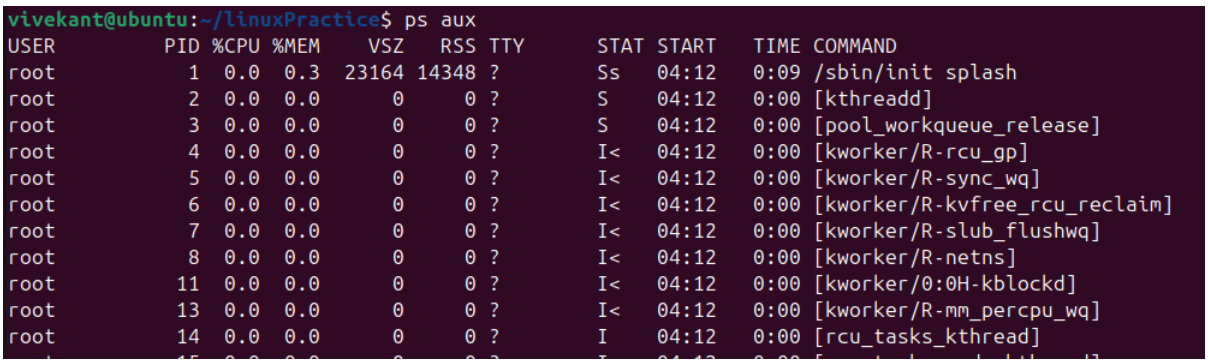
-
- syntax for sed

```
sed 'script' app.log
```

- examples
  - sed '/INFO/p' app.log
    - its suppose to pring line which contains 'INFO' but its pring all of the line so we have to put stric condition by placing '-n' before script
  - sed -n '/INFO/p' app.log
  - sed 's/INFO/LOG/g' app.log
    - it will replace all 'INFO' substring in file by LOG globaly
    - s stand for substitute , g stand for globally , so substitute globaly.
  - sed -n -e '/INFO/=' app.log
    - it will print line number which contains 'INFO'
    - -e for expression , here we used expression '='
    - '=' stand for line number
  - sed -n -e '/INFO/=' -e '/INFO/p' app.log
    - it will print line number with corresponding line which contains 'INFO' word as substring
  - sed '1,20 s/INFO/LOG/g' app.log
    - it will replace INFO substring by LOG from line 1 to lin 20
  - sed '1,20 s/INFO/LOG/g; 1,20p;21q' app.log
    - (1,20 s/INFO/LOG/g;)---> it will substitute 'INFO' by 'LOG' from line 1 to 20

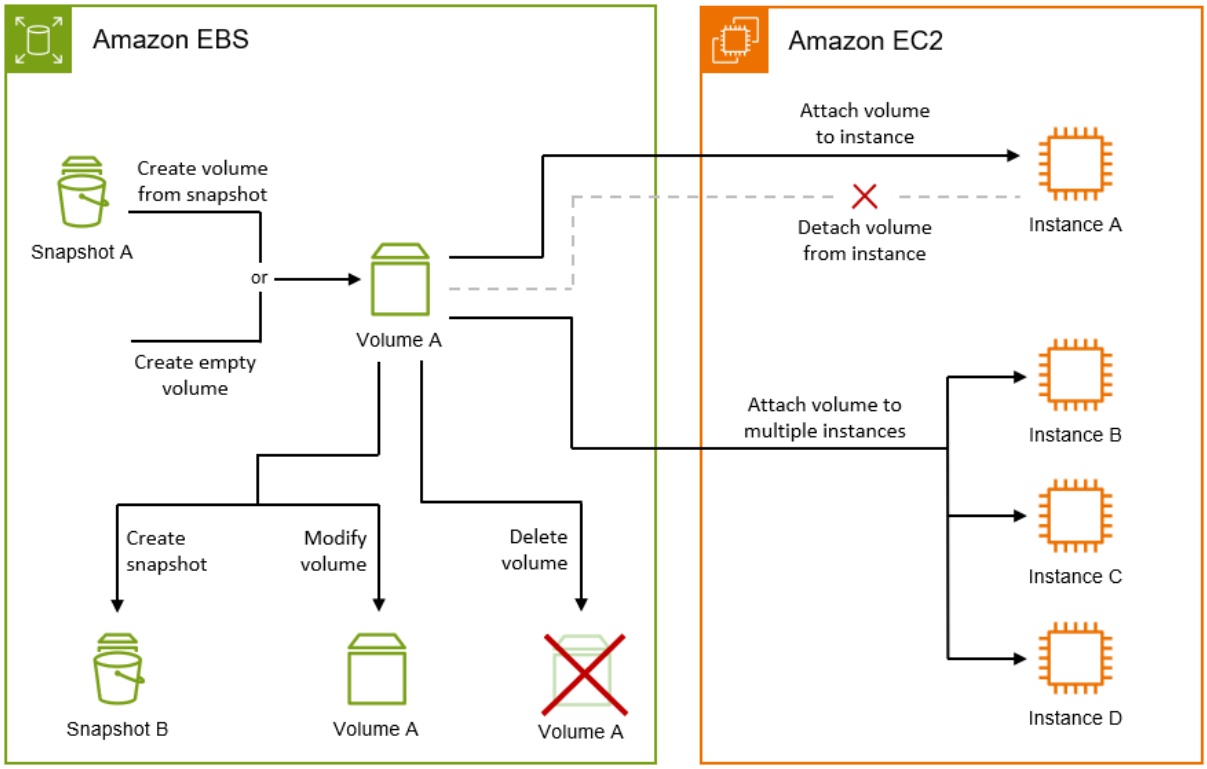
- 1,20p---> it is responsible for printing line 1 to 20
- 21q ---> program will quit at line 21
- ';' is for scritp end , then you can write new script

GREP(global regulation expression pattern)

- grep INFO app.log - it will give all lines of log which contains 'INFO'
- grep -i info app.log
  - -i for case insensitive, so it will print all line which contains info in any form capital or small
- grep -i -c info app.log
  - it will give count of info in app.log
- ps aux | grep vivekant
  - it will give all process which started by user="vivekant"
  - 
  - 
- ps aux | grep vivekant | awk '{print $2}'
  - it will give all process's PID which started by user="vivekant"
- ps aux | grep vivekant | awk '{print $2}' | awk '{count++} END {print count}'
  - it will give you total count of process which started by vivekant

Volume

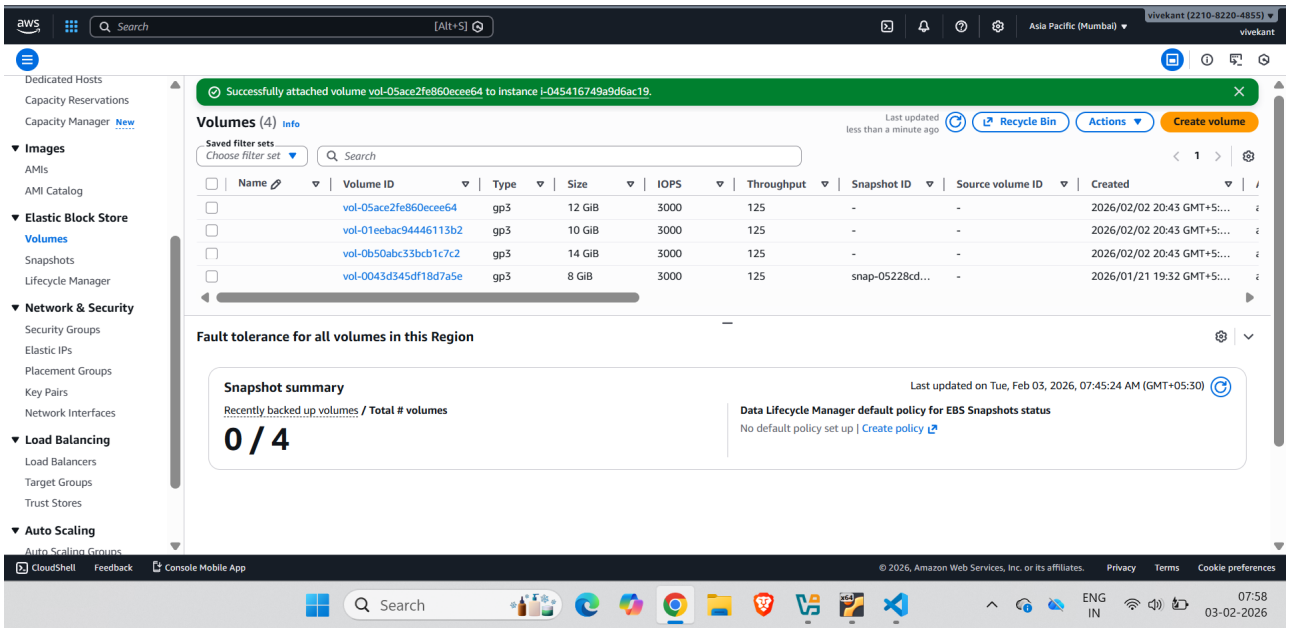
- in window we called block or disk which is name for storage, in linux we call volume thats its nothing special
- aws EBS
  - EBS ka full form hota hai Elastic Block Store — ye ek cloud hard disk jaisa hota hai jo servers ke saath attach hota hai (especially AWS servers).



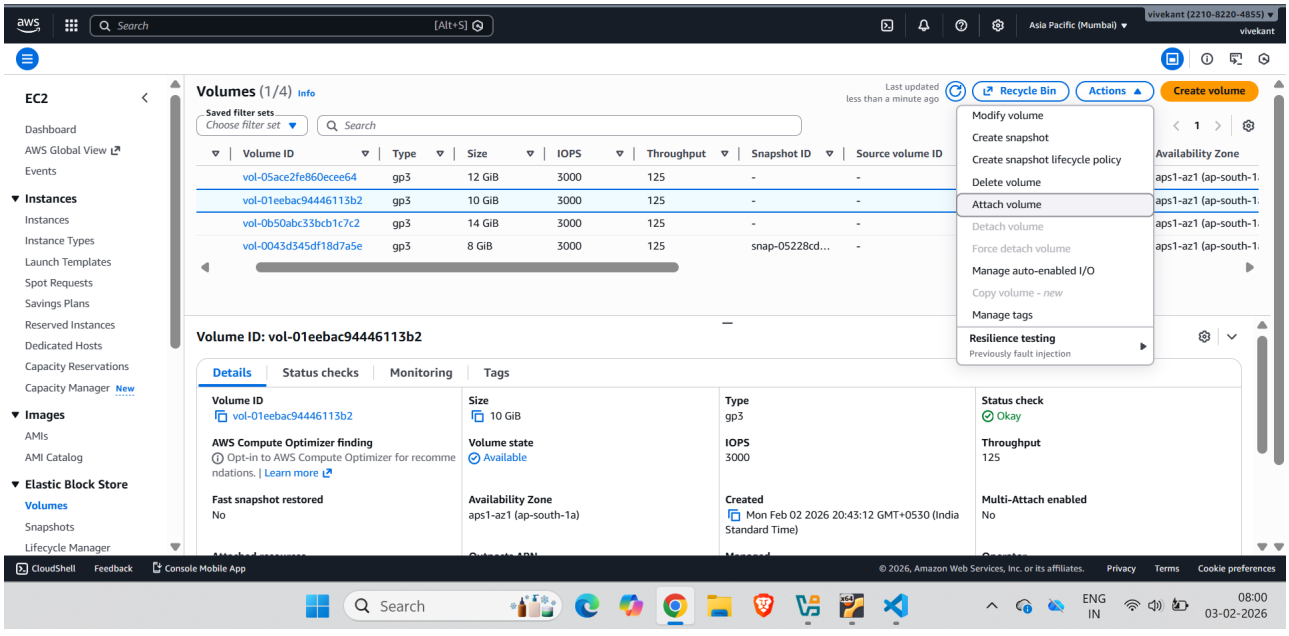
- lsblk
  - will list all block details (total size, occupied , unoccupied)

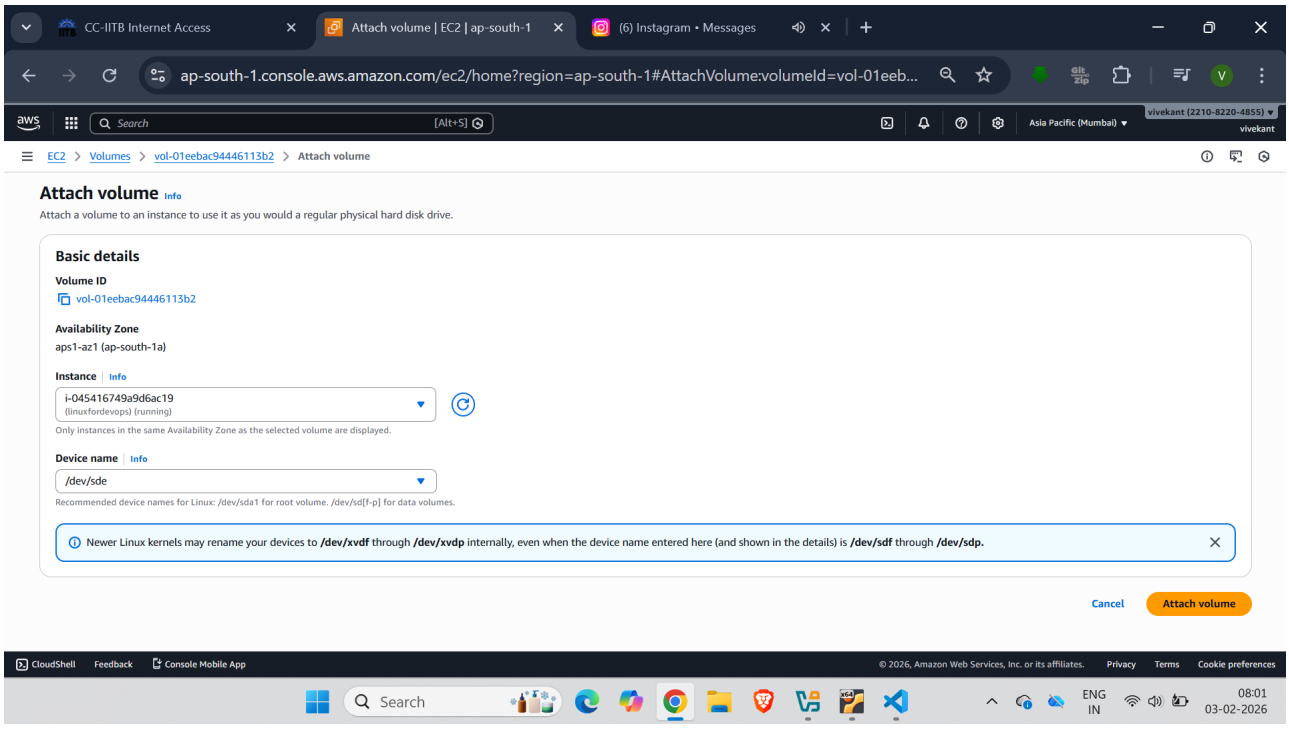
```
ubuntu@ip-172-31-47-102:~$ lsblk
NAME     MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0      7:0    0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1      7:1    0 27.8M  1 loop /snap/amazon-ssm-agent/12322
loop2      7:2    0 73.9M  1 loop /snap/core22/2216
loop3      7:3    0   74M  1 loop /snap/core22/2292
loop4      7:4    0 50.9M  1 loop /snap/snapd/25577
loop5      7:5    0 48.1M  1 loop /snap/snapd/25935
xvda     202:0    0    8G  0 disk
├─xvda1  202:1    0    7G  0 part /
├─xvda14 202:14   0    4M  0 part
├─xvda15 202:15   0  106M  0 part /boot/efi
└─xvda16 259:0    0  913M  0 part /boot
ubuntu@ip-172-31-47-102:~$
```

- mouting a volume means giving a path to volume
- attaching a volume mean attaching to ec2 instance
- attaching doesnt implies mounting
- create volume on aws

- 

- attach volumne to you instance(choose any instance that you have created)

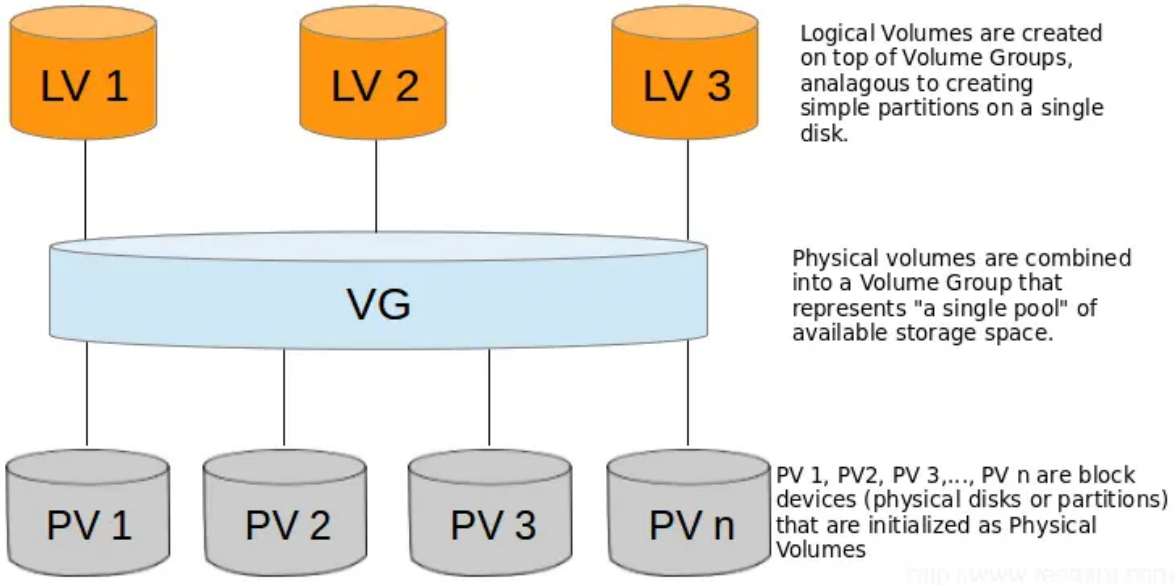
- 

- 

- device name is just like we use in our pc C , D , M disk
- logical volume is same as we do partition of our storage in pc, physical volume are volume that are available which we created on aws via ssh connection throught lvm, volume group are combined volume of all physical volume, the volume we are created on aws are just disk/block not physical volume(after disk/blakc we have to creat physical volume from them)

```
ubuntu@ip-172-31-47-102:~$ sudo su
root@ip-172-31-47-102:/home/ubuntu# lvm
lvm> exit
  Exiting.
root@ip-172-31-47-102:/home/ubuntu# lsblk
NAME     MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0      7:0    0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1      7:1    0 27.8M  1 loop /snap/amazon-ssm-agent/12322
loop2      7:2    0 73.9M  1 loop /snap/core22/2216
loop3      7:3    0   74M  1 loop /snap/core22/2292
loop4      7:4    0 50.9M  1 loop /snap/snapd/25577
loop5      7:5    0 48.1M  1 loop /snap/snapd/25935
xvda     202:0    0    8G  0 disk
├─xvda1  202:1    0    7G  0 part /
├─xvda14 202:14   0    4M  0 part
├─xvda15 202:15   0  106M  0 part /boot/efi
└─xvda16 259:0    0  913M  0 part /boot
xvdd     202:48   0   14G  0 disk
xvde     202:64   0   10G  0 disk
xvdf     202:80   0   12G  0 disk
root@ip-172-31-47-102:/home/ubuntu#
```

  - these all thing managed by LVM(logical volume management) -



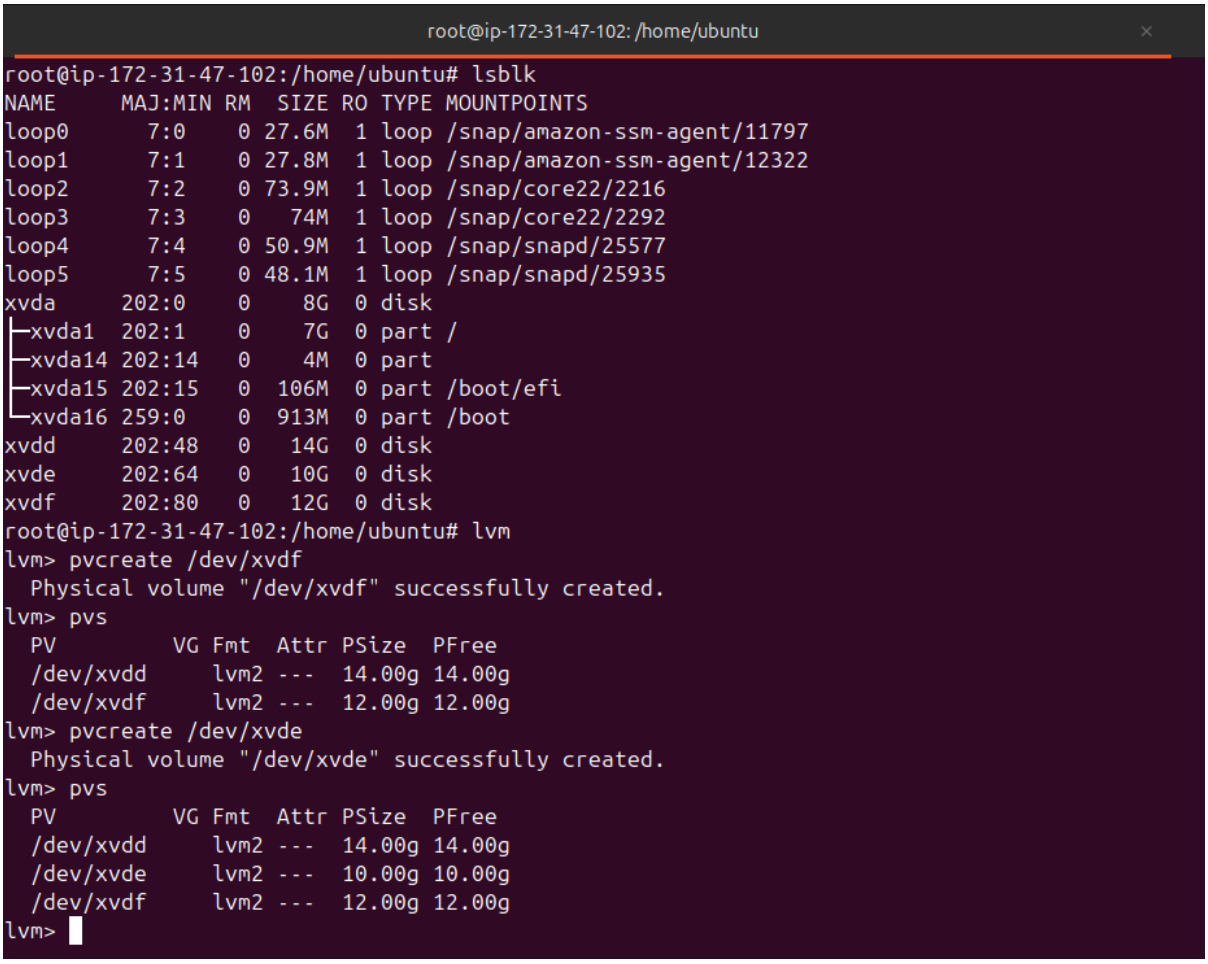
- sudo su
- this command will make you as root user so you dont need to use sudo again and again

lvm

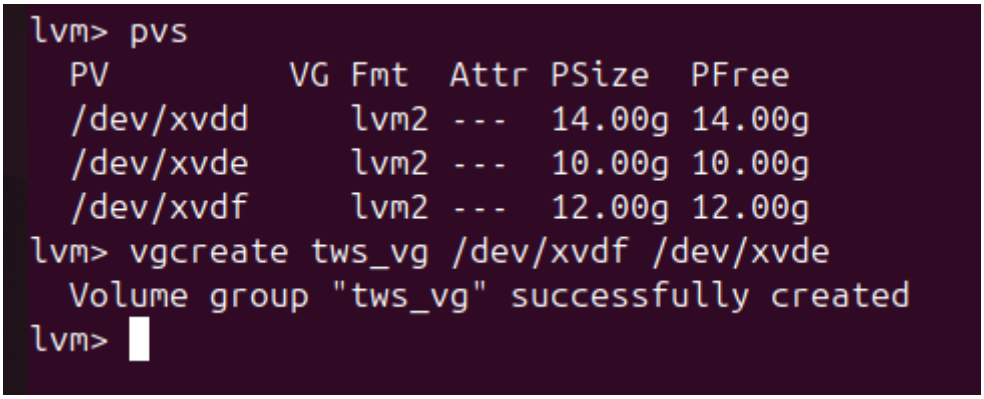
- first switch to root user so you dont have to use sudo again and again

physical volume creation from disk available

- pvcreate disk_name

- it will create physical volume from disk you have created on aws
- 

volume group creation from available physical volume(created above)

- vgcreate volume_group_name pv1 pv2
  - it will create volume group by combining physical volume1 and physical volume2
  - 

- vgs
- it will tell you all available volume group
- pvs
- it will tell you all available physical volumes

creating logical volume from volume group as much we required

- lvcreate -L 10G -n tws_lv1 tws_vg
  - it will create 10GB logical volume from volume group tws_vg
  - '-L' for size , '-n' for name you want for logical volume

```
lvm> vgs
  VG     #PV #LV #SN Attr   VSize  VFree
  tws_vg   2   0   0 wz--n- 21.99g 21.99g
lvm> lvcreate -L 10G -n tws_lv1 tws_vg
  Logical volume "tws_lv1" created.
lvm> lvs
  LV      VG     Attr       LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  tws_lv1 tws_vg -wi-a----- 10.00g
lvm>
```

- lvs give you list of logical volume with their volume_group name

pvdisplay

- it will give you details of each of available physical volumes

```
lvm> pvdisplay
  --- Physical volume ---
  PV Name               /dev/xvdf
  VG Name               tws_vg
  PV Size               12.00 GiB / not usable 4.00 MiB
  Allocatable           yes
  PE Size               4.00 MiB
  Total PE              3071
  Free PE               511
  Allocated PE          2560
  PV UUID               dulGlx-F6FK-u3fv-GZFS-FQAc-TKbQ-uvQfUb

  --- Physical volume ---
  PV Name               /dev/xvde
  VG Name               tws_vg
  PV Size               10.00 GiB / not usable 4.00 MiB
  Allocatable           yes
  PE Size               4.00 MiB
  Total PE              2559
  Free PE               2559
  Allocated PE          0
  PV UUID               CjKQtR-ADsd-FcmR-5Z9f-MszK-IeZP-a1bUsy

  "/dev/xvdd" is a new physical volume of "14.00 GiB"
  --- NEW Physical volume ---
  PV Name               /dev/xvdd
  VG Name
  PV Size               14.00 GiB
  Allocatable           NO
  PE Size               0
  Total PE              0
  Free PE               0
  Allocated PE          0
```

vgdisplay

- it will give you details of each of available volume groups

```
lvm> vgdisplay
  --- Volume group ---
  VG Name               tws_vg
  System ID
  Format                lvm2
  Metadata Areas        2
  Metadata Sequence No  2
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                1
  Open LV               0
  Max PV                0
  Cur PV                2
  Act PV                2
  VG Size               21.99 GiB
  PE Size               4.00 MiB
  Total PE              5630
  Alloc PE / Size       2560 / 10.00 GiB
  Free  PE / Size       3070 / 11.99 GiB
  VG UUID               j90f8x-L6jX-2jPH-h3Jl-35H9-twH7-54SDKN

lvm>
```

mkfs.ext4 /dev/volume_group_name/logical_volume_name

- example mkfs.ext4 /dev/tws_vg/tws_lv1
- it is use to formate logical volumn
- before mounting we formate logical volume

mounting logical volume

- command: mount logical_volume_path path_where_you_want_to_mount_the_volume

```
root@ip-172-31-47-102:/# ls mnt
tws_lv1_mount
root@ip-172-31-47-102:/# mount /dev/tws_vg/tws_lv1 /mnt/tws_lv1_mount
root@ip-172-31-47-102:/# df -h
Filesystem                 Size  Used Avail Use% Mounted on
/dev/root                  6.8G  3.7G  3.1G  55% /
tmpfs                      479M     0  479M   0% /dev/shm
tmpfs                      192M  940K  191M   1% /run
tmpfs                      5.0M     0  5.0M   0% /run/lock
/dev/xvda16                881M   89M  730M  11% /boot
/dev/xvda15                105M  6.2M   99M   6% /boot/efi
tmpfs                       96M   16K   96M   1% /run/user/1000
/dev/mapper/tws_vg-tws_lv1  9.8G   24K  9.3G   1% /mnt/tws_lv1_mount
root@ip-172-31-47-102:/#
```

umount path_where_you_mounted_the_volume

- command: umount mnt/tws_lv1_mount(path where you mount the volume)
- on unmounting , data in that volume are not going to delete
- the only change will you see is you cant read file inside unmounted volume

```
root@ip-172-31-47-102:/mnt/tws_lv1_mount# cd ../..
root@ip-172-31-47-102:/# cat mnt/tws_lv1_mount/devops/hello-dosto.txt
i love dhara sachin shah truely
if she agree then i am ok to marry her

root@ip-172-31-47-102:/# unmount mnt/tws_lv1_mount/
Command 'unmount' not found, did you mean:
  command 'umount' from deb mount (2.39.3-9ubuntu6.4)
Try: apt install <deb name>
root@ip-172-31-47-102:/# umount mnt/tws_lv1_mount/
root@ip-172-31-47-102:/# cat mnt/tws_lv1_mount/devops/hello-dosto.txt
cat: mnt/tws_lv1_mount/devops/hello-dosto.txt: No such file or directory
root@ip-172-31-47-102:/# mount dev/tws_vg/tws_lv1 mnt/tws_lv1_mount/
root@ip-172-31-47-102:/# cat mnt/tws_lv1_mount/devops/hello-dosto.txt
i love dhara sachin shah truely
if she agree then i am ok to marry her

root@ip-172-31-47-102:/#
```

mounting logical volume

1. we create volume group of physical volume
2. then we make logical volume from volume group as our desire size
3. the we do formating of logical volume
4. then we do mounting

mounting directely physical volume

1. we formate physical volume
2. then mount it
3. formate command for physical volume/disk-->mkfs -t ext4 dev/xvdd (physical volume path)
4. mounting command will be same : mount physical_volume_path path_where_you_want_to_mount_physical_volume

Dynamic storage management using LVM with aws EBS(elastic block storage)

how to inc size of logical volume

- syntax: lvextend -L +5G logical_volume_path(not that path where you mounted the logical volume like in our case i mounted it in /mnt/tws_lv1_mount)
- size will inc from volume_group from which the logical volume were created
- command: lvextend -L +5G /dev/tws_vg/tws_lv1

```
root@ip-172-31-47-102:/dev# lvextend -L +5G /dev/tws_vg/tws_lv1
  Size of logical volume tws_vg/tws_lv1 changed from 10.00 GiB (2560 extents) to 15.00 GiB (3840 extents).
  Logical volume tws_vg/tws_lv1 successfully resized.
root@ip-172-31-47-102:/dev#
```

- you can see the change by runnign lsblk command

```
root@ip-172-31-47-102:/dev# lsblk
NAME             MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0              7:0    0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1              7:1    0 27.8M  1 loop /snap/amazon-ssm-agent/12322
loop2              7:2    0 73.9M  1 loop /snap/core22/2216
loop3              7:3    0   74M  1 loop /snap/core22/2292
loop4              7:4    0 50.9M  1 loop /snap/snapd/25577
loop5              7:5    0 48.1M  1 loop /snap/snapd/25935
xvda             202:0    0    8G  0 disk
├─xvda1          202:1    0    7G  0 part /
├─xvda14         202:14   0    4M  0 part
├─xvda15         202:15   0  106M  0 part /boot/efi
└─xvda16         259:0    0  913M  0 part /boot
xvdd             202:48   0   14G  0 disk /mnt/tws_disk_mount
xvde             202:64   0   10G  0 disk
└─tws_vg-tws_lv1 252:0    0   15G  0 lvm  /mnt/tws_lv1_mount
xvdf             202:80   0   12G  0 disk
└─tws_vg-tws_lv1 252:0    0   15G  0 lvm  /mnt/tws_lv1_mount
root@ip-172-31-47-102:/dev#
```

what is difference between attached and mount volume

- attached volume is added to instance but you cant use that
- mounted volume is usable
- so to make volume usable you have to mount it first

awk

- it only work on CSV(comman seprated value) or TSV(tab/space seprated value) data
- for any kind of file , we use sed(stream editor)
- syntax for awk will be

```
awk 'script' file
```

- example
  - awk '/INFO/ {count++} END {print "The number of line which contains 'INFO' = " count}' app.log
    - it will print how many line which contains 'INFO'
  - awk '$1=="[2010-04-24" {print}' app.log
    - it will seprate all the log of date 2010-04-24

```
ubuntu@ip-172-31-47-102:~/awk$ awk '$1=="[2010-04-24" {print}' app.log
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): Execution begin.
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all options
[2010-04-24 07:51:54,393]  INFO - [main] Attempting to log in...
[2010-04-24 07:51:55,081]  INFO - [main] Successfully logged in as: wchung/eric
[2010-04-24 07:51:55,081] DEBUG - [main] BulkOpsClient.doImport(): Execution begin.
[2010-04-24 07:51:55,081]  INFO - [main] Validating Oracle Data Loader On Demand
[2010-04-24 07:51:55,081] DEBUG - [main] FieldMappingManager.parseMappings():
[2010-04-24 07:51:55,097] DEBUG - [main] FieldMappingManager.parseMappings():
```

  - awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,1000]" {print}' app.log
    - it will print all log of date 2010-04-24 and time 7:51:00,000 to 7:52:00,000

```
ubuntu@ip-172-31-47-102:~/awk$ awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,1000]" {print}' app.log
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): Execution begin.
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all
[2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all options
[2010-04-24 07:51:54,393]  INFO - [main] Attempting to log in...
[2010-04-24 07:51:55,081]  INFO - [main] Successfully logged in as: wchung/eric
[2010-04-24 07:51:55,081] DEBUG - [main] BulkOpsClient.doImport(): Execution begin.
[2010-04-24 07:51:55,081]  INFO - [main] Validating Oracle Data Loader On Demand
[2010-04-24 07:51:55,081] DEBUG - [main] FieldMappingManager.parseMappings():
[2010-04-24 07:51:55,097] DEBUG - [main] FieldMappingManager.parseMappings():
[2010-04-24 07:51:55,331] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run():
[2010-04-24 07:51:55,331]  INFO - [main] A SOAP request was sent to the server to
[2010-04-24 07:51:55,862] DEBUG - [Thread-1]
[2010-04-24 07:51:55,862] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run():
[2010-04-24 07:51:55,862] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run(): SOAP
[2010-04-24 07:51:55,862] DEBUG - [Thread-1] ODWSSessionKeeperThread.Run(): Going
[2010-04-24 07:51:55,862] DEBUG - [main]
[2010-04-24 07:51:55,862] DEBUG - [main] There was an error sending the SOAP request
[2010-04-24 07:51:55,862] DEBUG - [main] BulkOpsClient.sendValidationRequest():
[2010-04-24 07:51:56,862]  INFO - [main] A SOAP request was sent to the server to
```

- awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,000]" {count++} END {print count}' app.log
  - it will give you total count of log line of desire date and time
  - millisecond shoudl be 000 to 999 if you put 1000 it will consider as hour:min:second+1,000
  - ```
    ubuntu@ip-172-31-47-102:~/awk$ awk '$1=="[2010-04-24" && $2>="07:51:00,000]" && $2<="07:52:00,000]" {count++} END {print count}' app.log
    19
    ```
- awk 'NR>=2 && NR<=10 {print}' app.log
  - it will give you second to 10th lines of log
  - ```
    ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10' app.log

    [2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all
    configurations loaded: {numofthreads=1, impstatchkinterval=30, maxloginattempts=1,
    manifestfiledir=.\Manifest\, sessionkeepchkinterval=300, routingurl=https://
    sso.crmondemand.com, hosturl=http://sdchs20n263.us.oracle.com, testmode=debug,
    maxthreadfailure=1, logintimeoutms=180000, csvblocksize=1000, maxsoapsize=10240}

    [2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all options
    loaded: {password=*********, clientloglevel=detailed, waitforcompletion=False,
    ```
- awk 'NR>=2 && NR<=10 {print NR}' app.log
- it will print row number from 2 to 10
- ```
  ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10 {print NR}' app.log
  2
  3
  4
  5
  6
  7
  8
  9
  10
  ```
- awk 'NR>=2 && NR<=10 {print NR, $0}' app.log
  - it will pring row number with corresponding log
  - $0 mean all column or whole line of a row
  - ```
    ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10 {print NR, $0}' app.log
    2
    3 [2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all
    4 configurations loaded: {numofthreads=1, impstatchkinterval=30, maxloginattempts=1,
    5 manifestfiledir=.\Manifest\, sessionkeepchkinterval=300, routingurl=https://
    6 sso.crmondemand.com, hosturl=http://sdchs20n263.us.oracle.com, testmode=debug,
    7 maxthreadfailure=1, logintimeoutms=180000, csvblocksize=1000, maxsoapsize=10240}
    8
    9 [2010-04-24 07:51:54,393] DEBUG - [main] BulkOpsClient.main(): List of all options
    10 loaded: {password=*********, clientloglevel=detailed, waitforcompletion=False,
    ubuntu@ip-172-31-47-102:~/awk$ awk 'NR>=2 && NR<=10 {print NR, $0}' app.log
    ```

linux cheatcode

- ctl+u remove all content from cursor to start
- '-' use before command is to give option to command.
- '-' ,Option = command ko extra instruction dena matlab kaise kaam karna hai and kis tool ka use karna hai